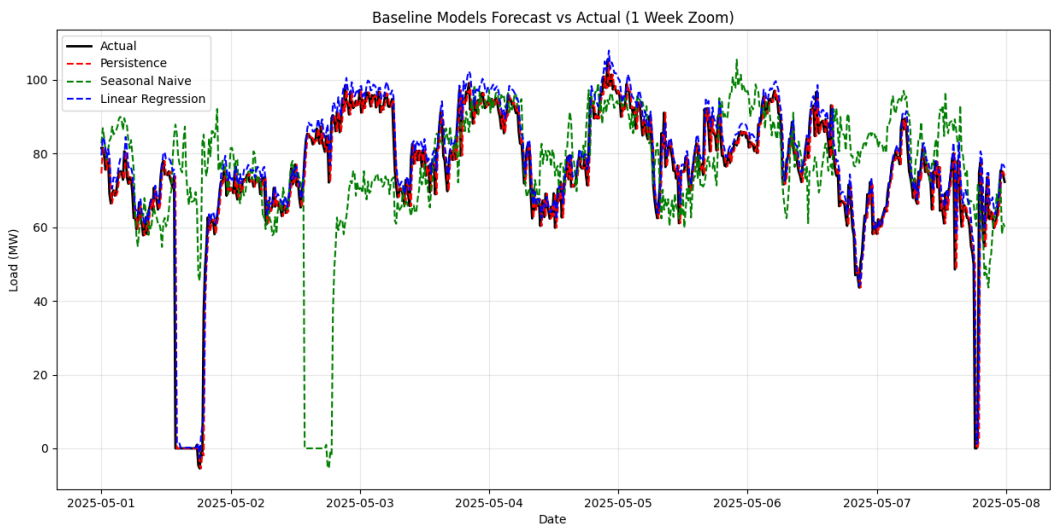


Load Forecasting: Plots Gallery

This document provides a visual index of all plots generated during the model building and analysis phases.

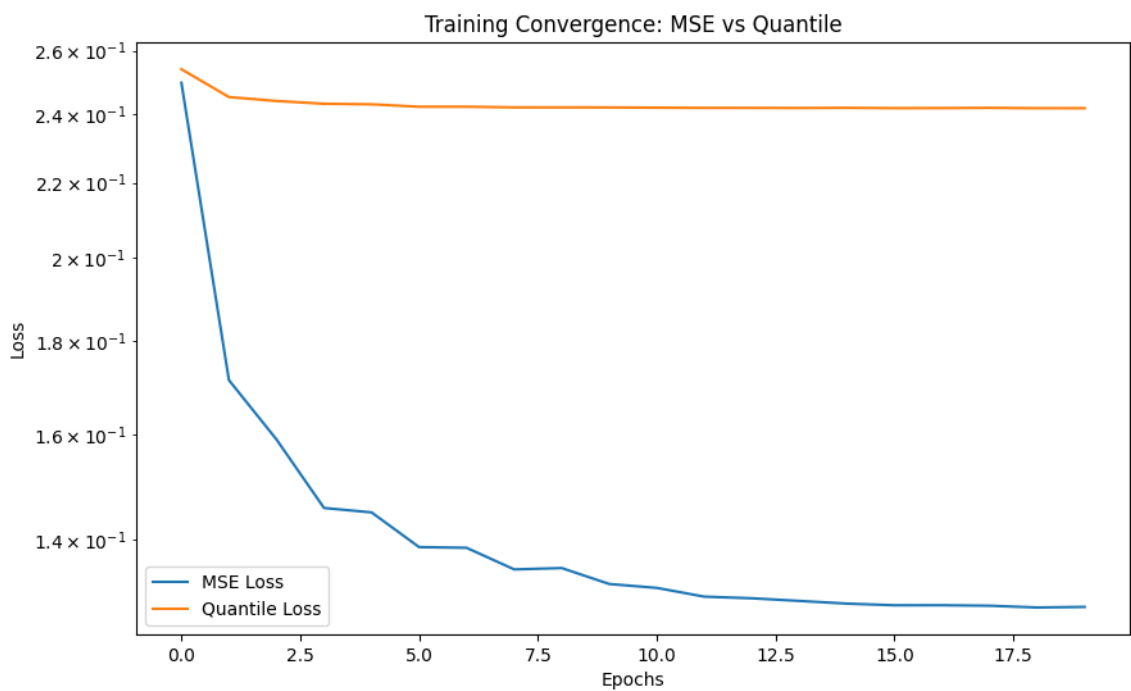
Root Plots

- baselines_forecast.png**



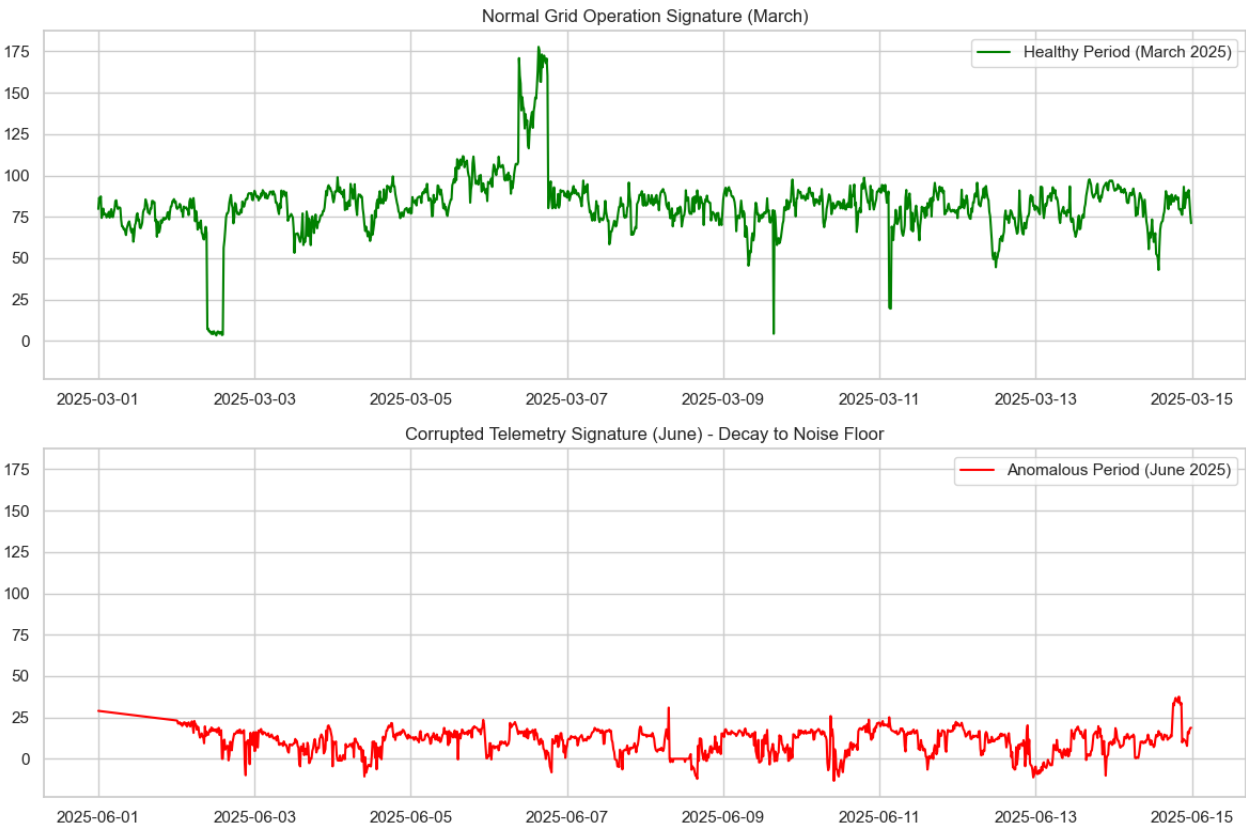
Ablation Study

- ablation/loss_convergence.png**



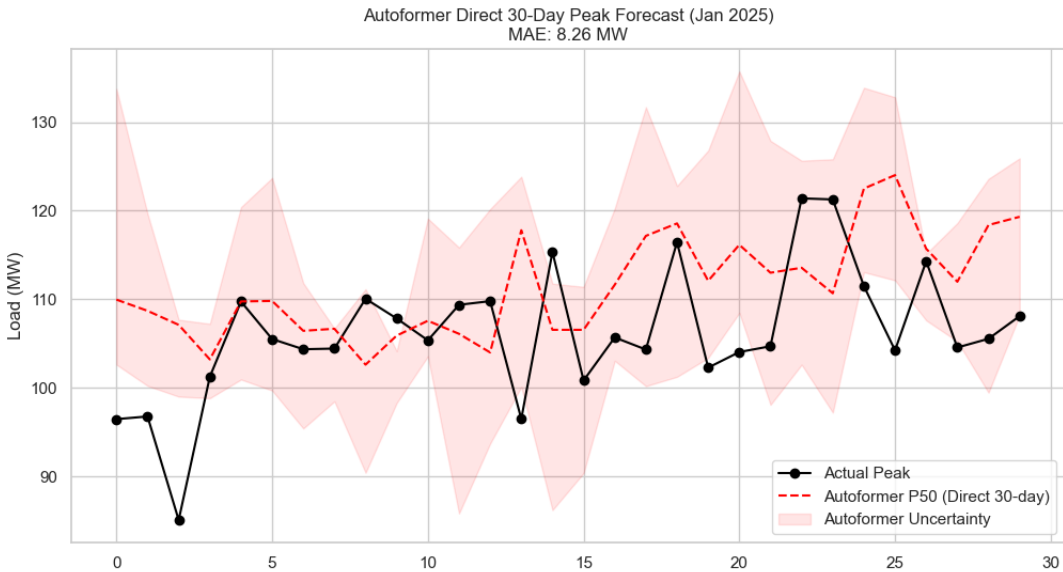
Anomaly Analysis

- anomaly_analysis/anomaly_comparison.png

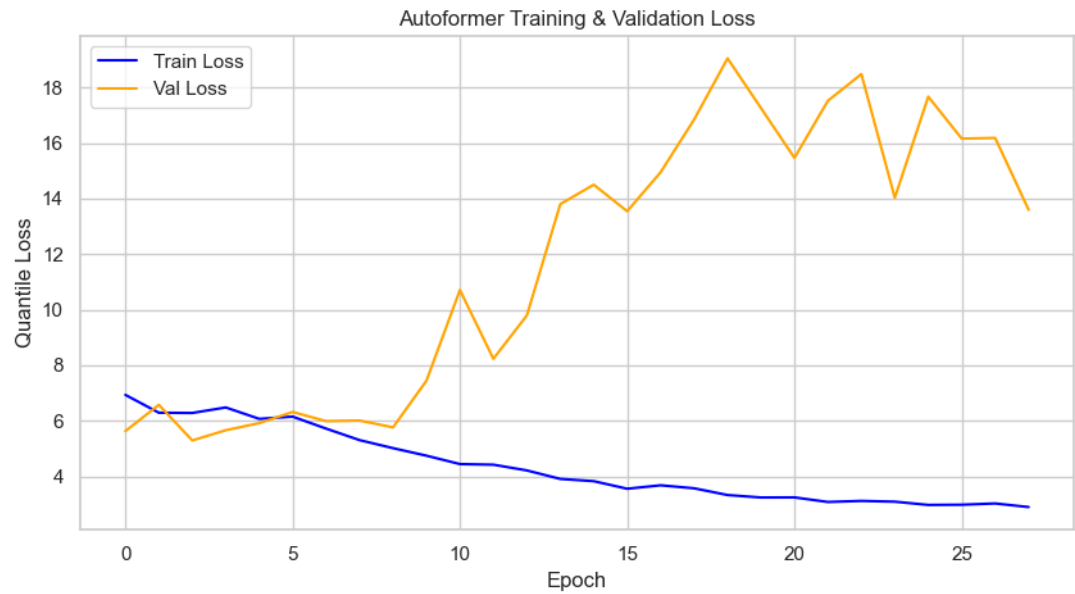


Autoformer Long-Term (LTLF)

- autoformer_ltlf/autoformer_vs_actual.png

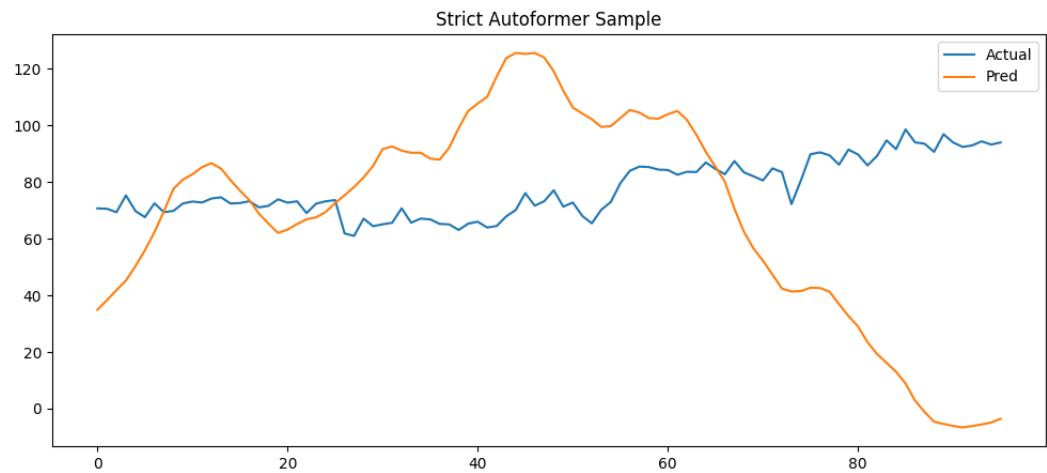


- **autoformer_itlf/loss_curves.png**



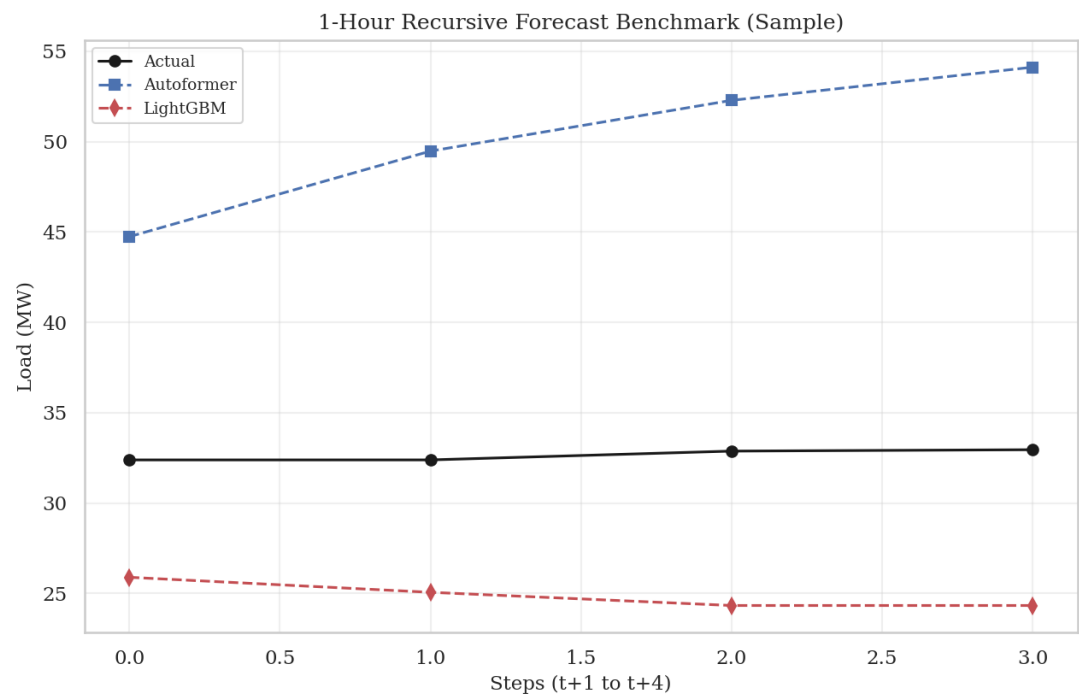
Autoformer Strict

- **autoformer_strict/autoformer_sample.png**



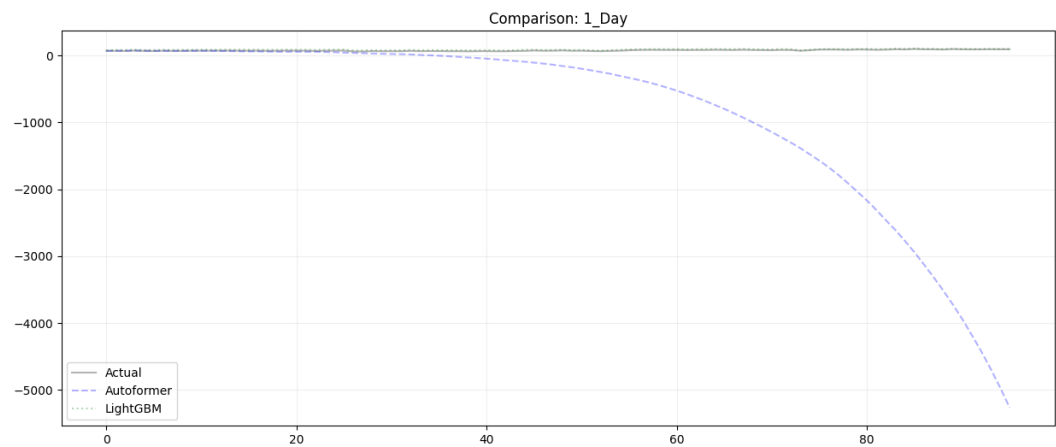
Comparison Benchmarks

- **comparison/benchmark_1hr_recursive.png**

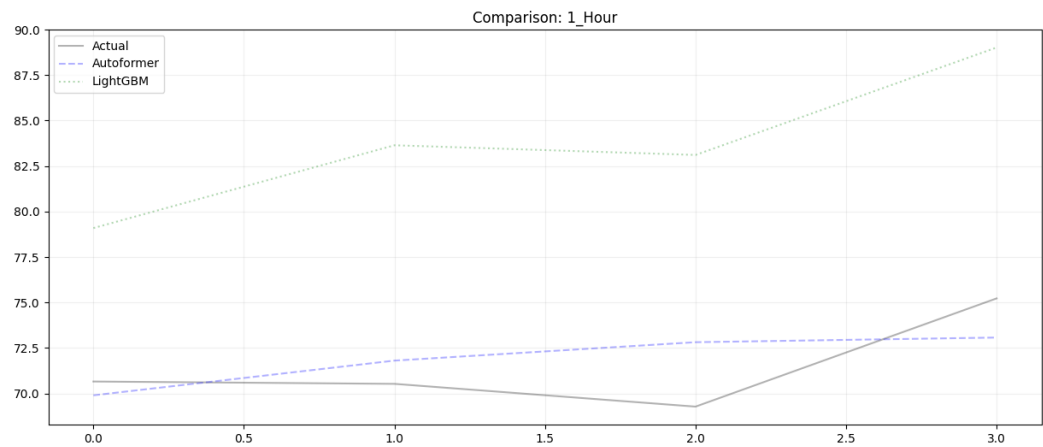


Comprehensive Strategy Analysis

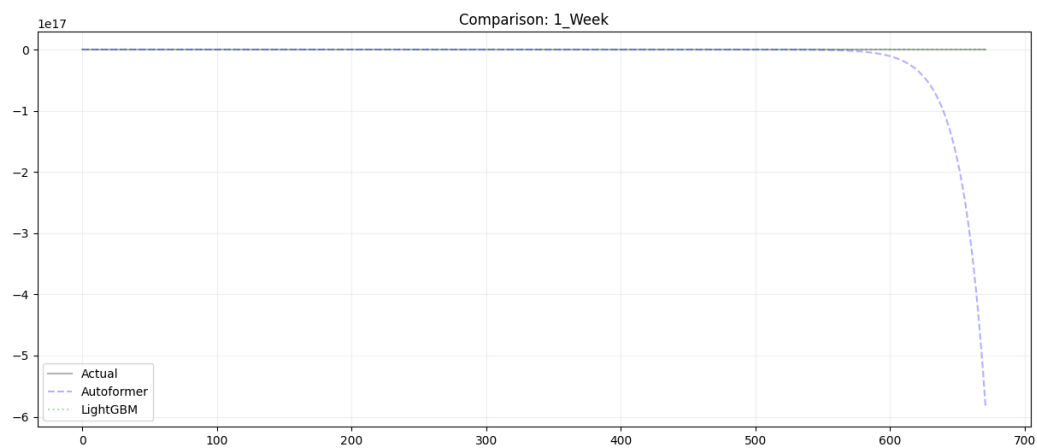
- **comprehensive/seasonal_1_Day.png**



- comprehensive/seasonal_1_Hour.png

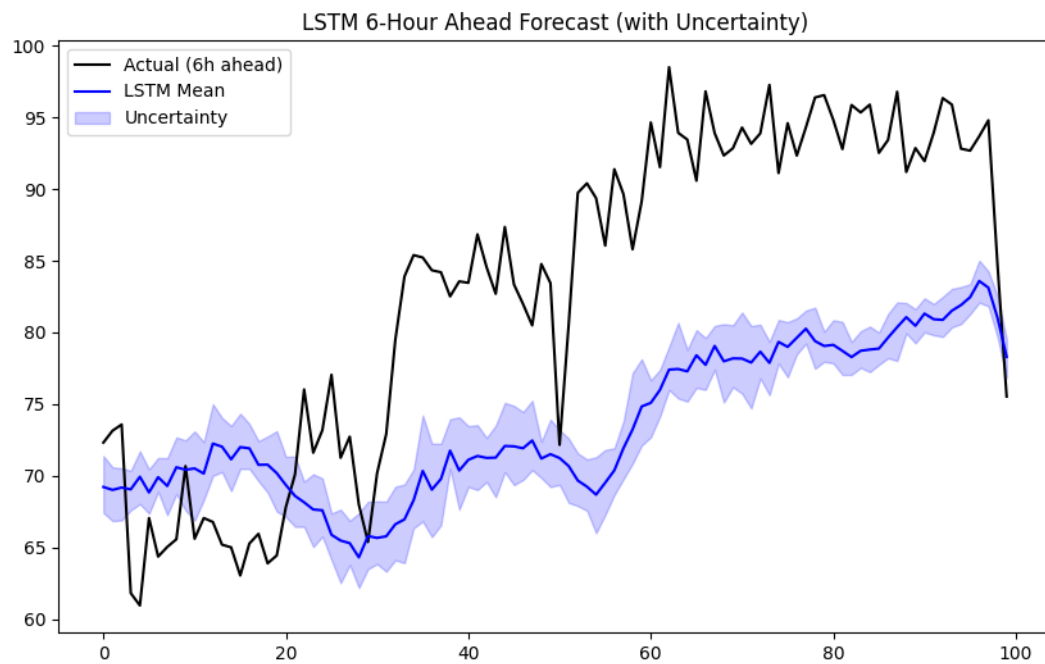


- comprehensive/seasonal_1_Week.png

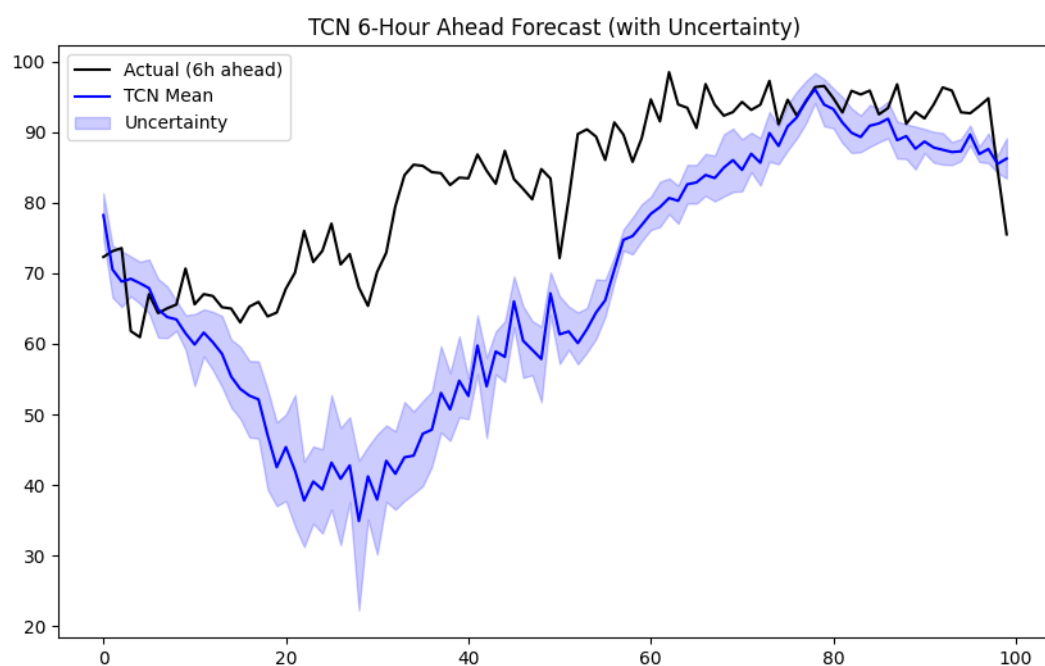


Deep Learning Comparisons

- **deep_learning/LSTM_6h_forecast.png**

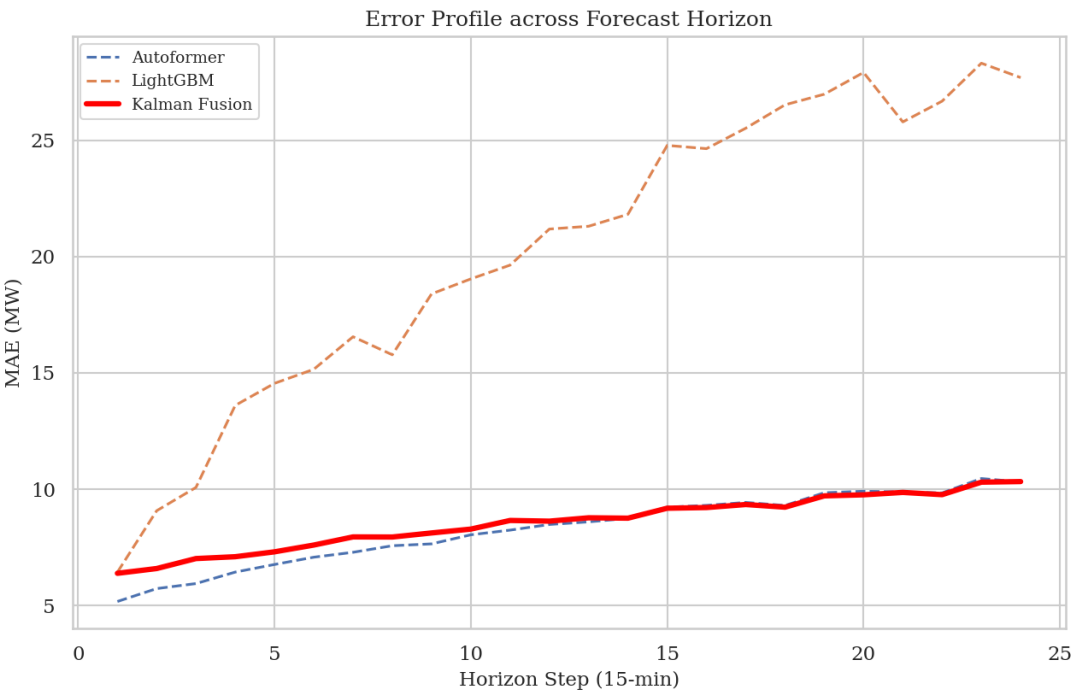


- **deep_learning/TCN_6h_forecast.png**

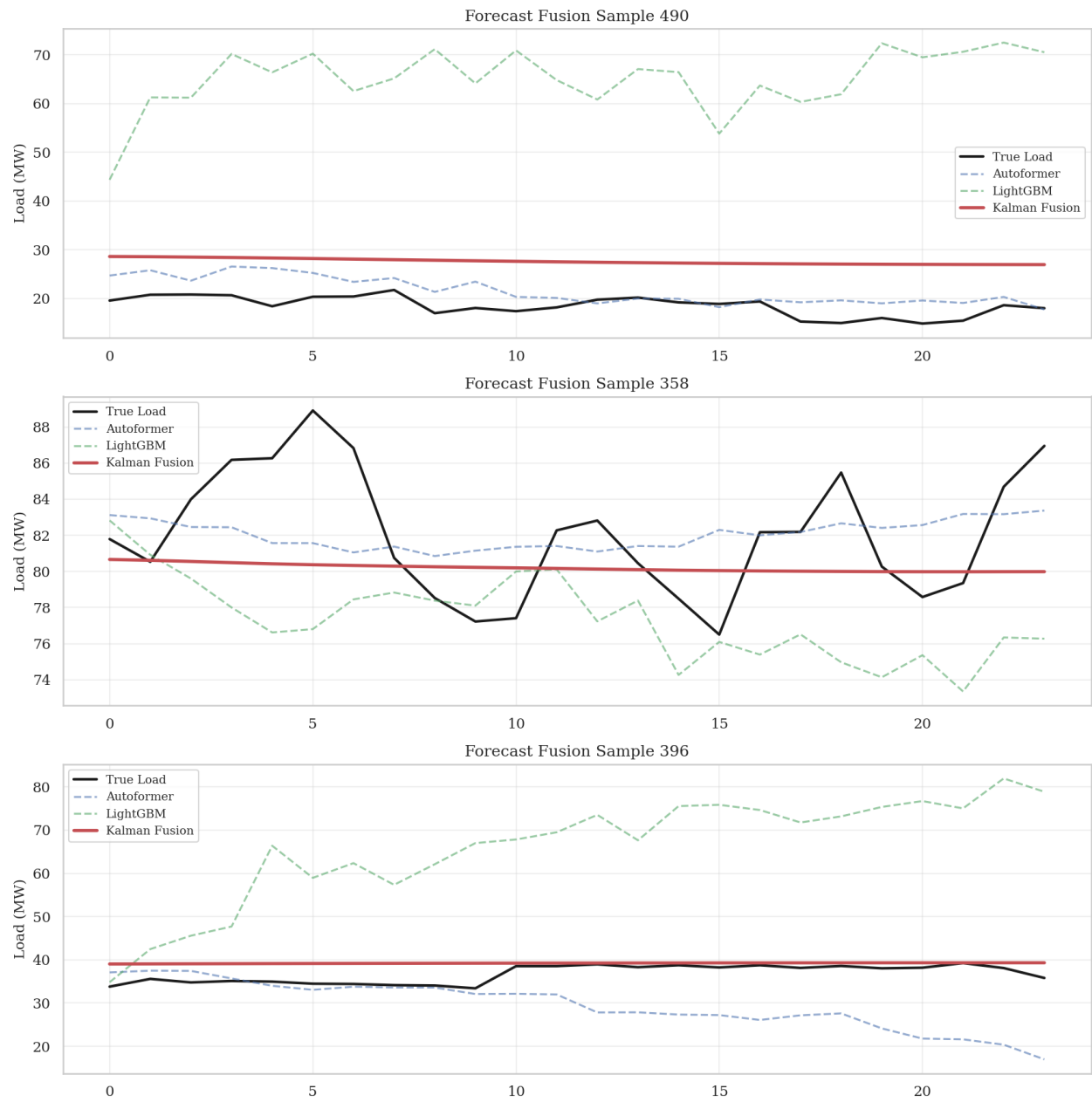


Ensemble (Kalman)

- ensemble/fusion_error_profile.png

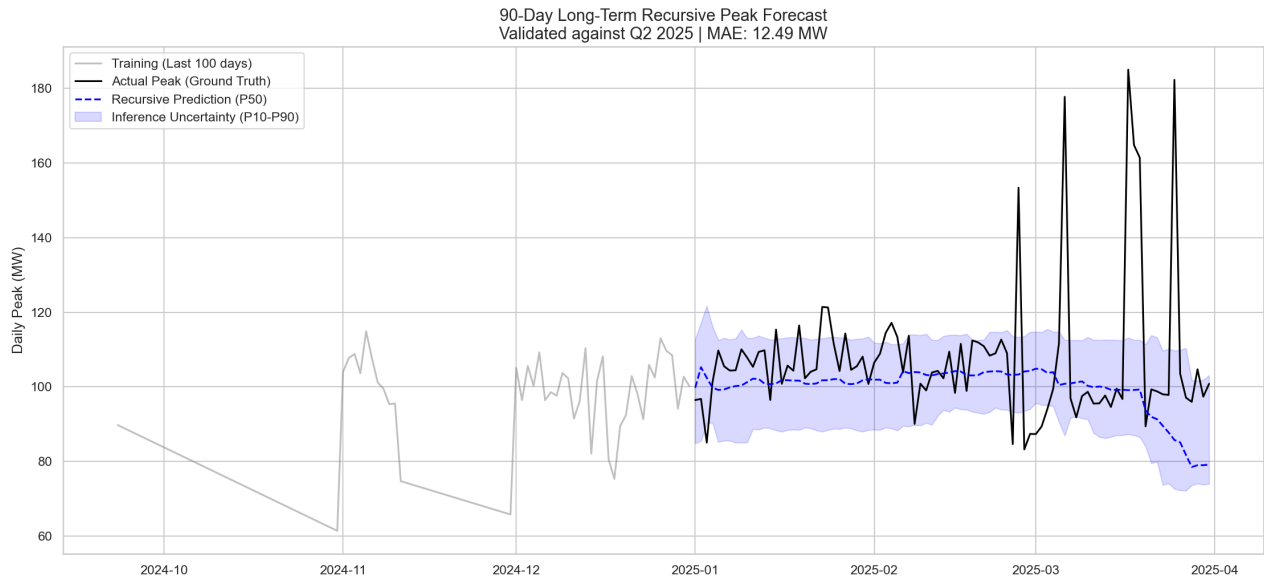


- ensemble/fusion_samples.png



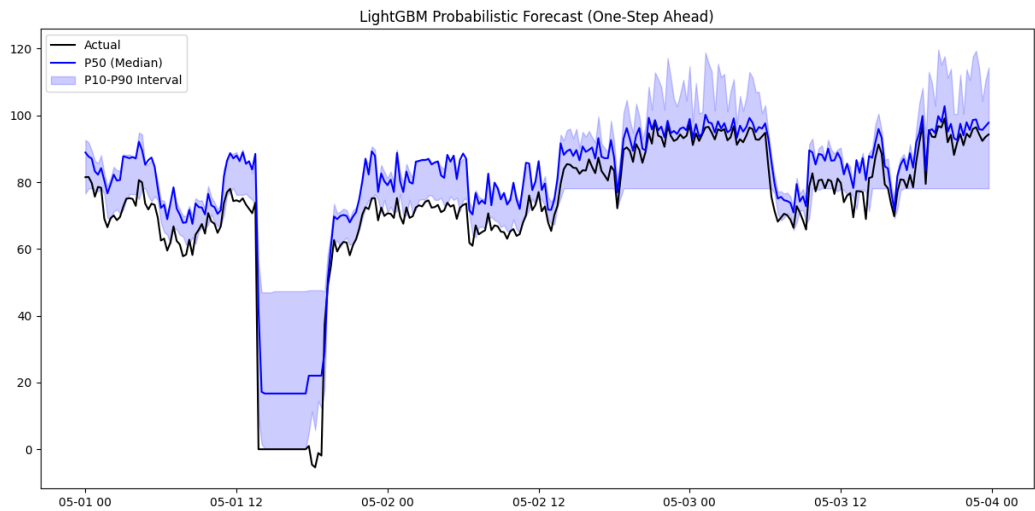
Final LTLF Rollout

- **final_ltlf/final_long_term_rollout.png**



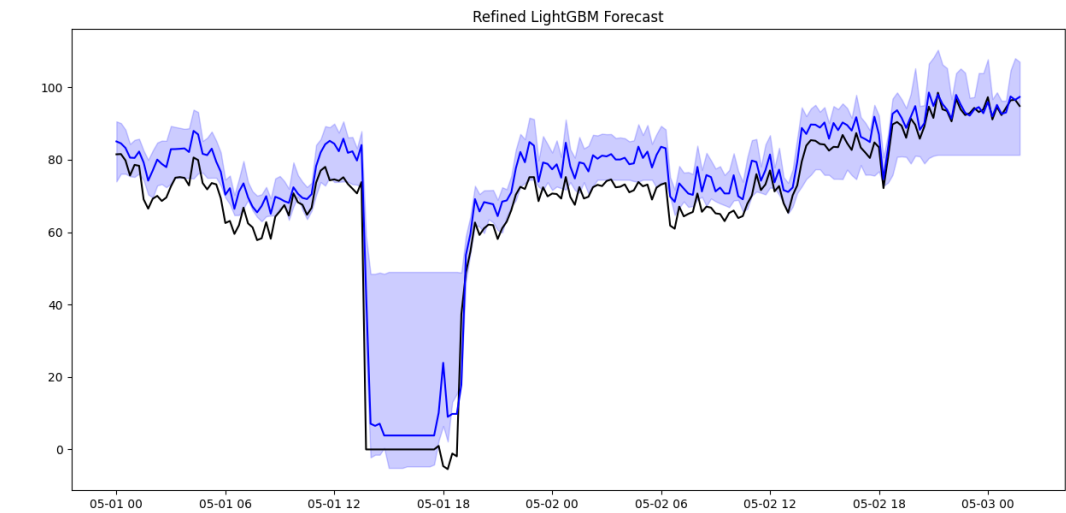
LightGBM Enhanced

- **lightgbm_enhanced/uncertainty_plot.png**



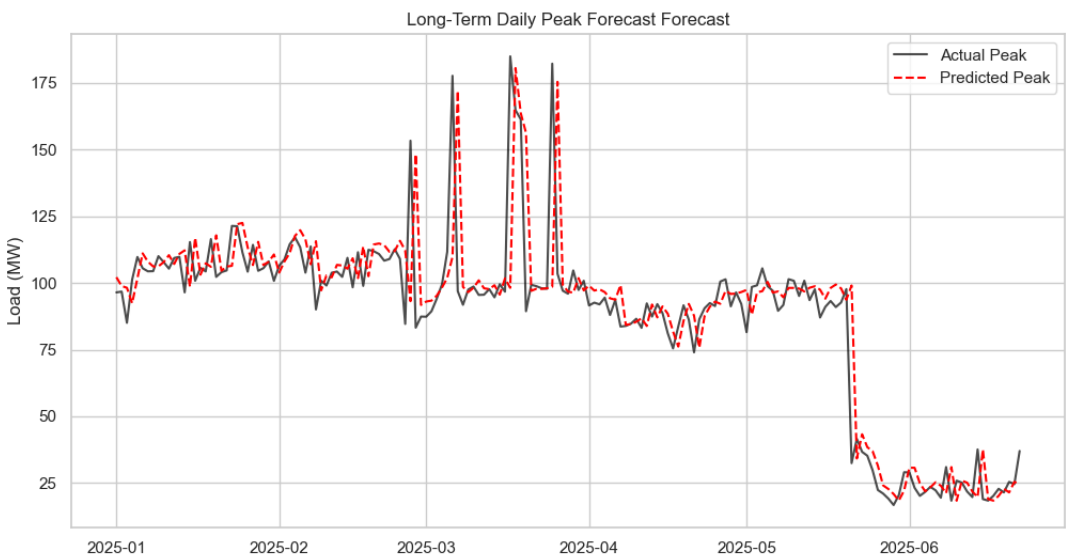
LightGBM Refined

- `lightgbm_refined/refined_forecast.png`

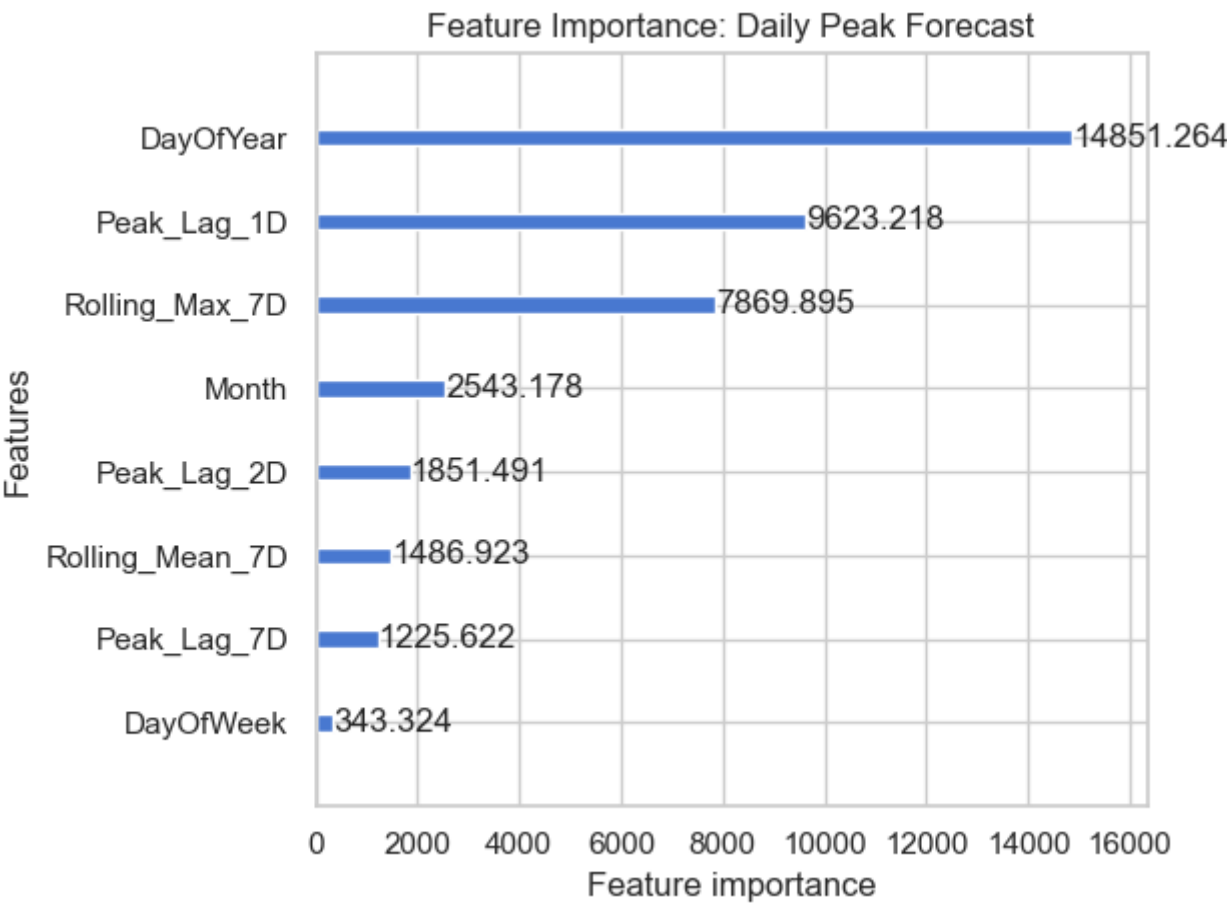


Long-Term Peak Forecasting

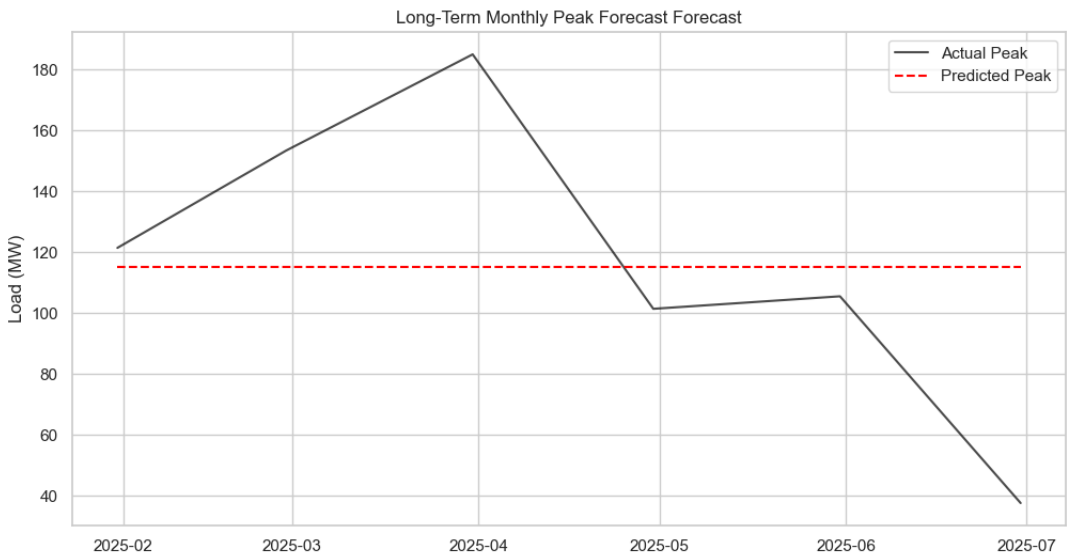
- `long_term_peaks/daily_peak_forecast_forecast.png`



- `long_term_peaks/daily_peak_forecast_importance.png`

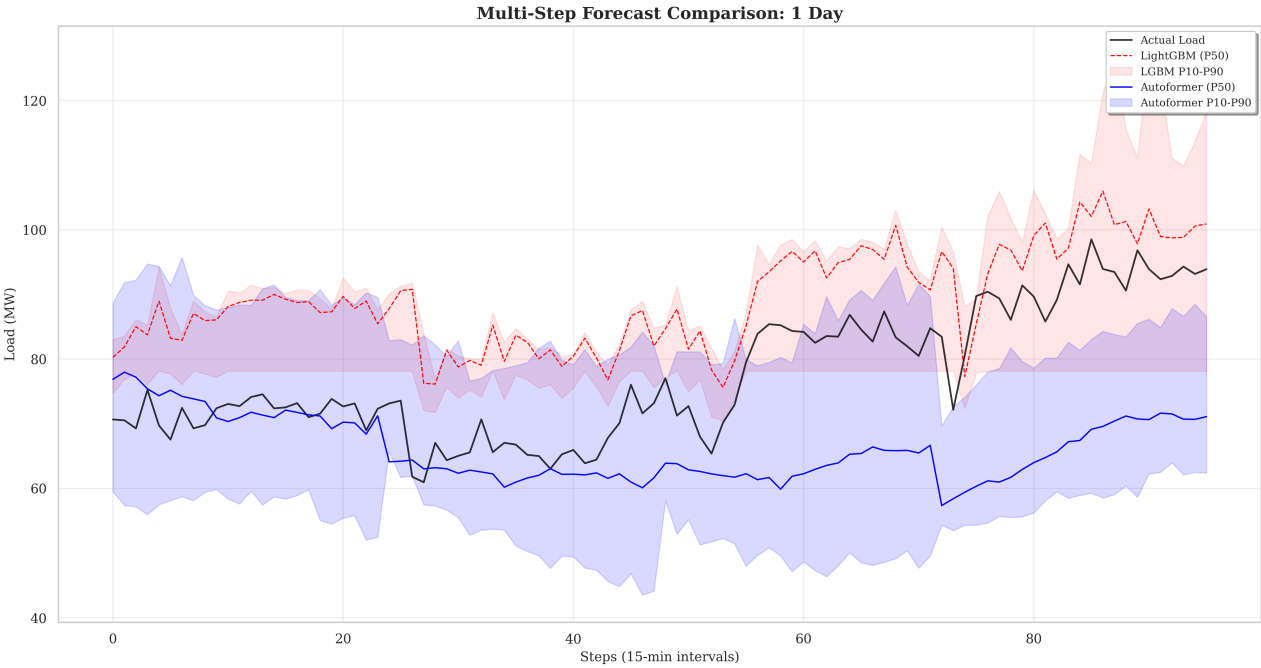


- `long_term_peaks/monthly_peak_forecast_forecast.png`

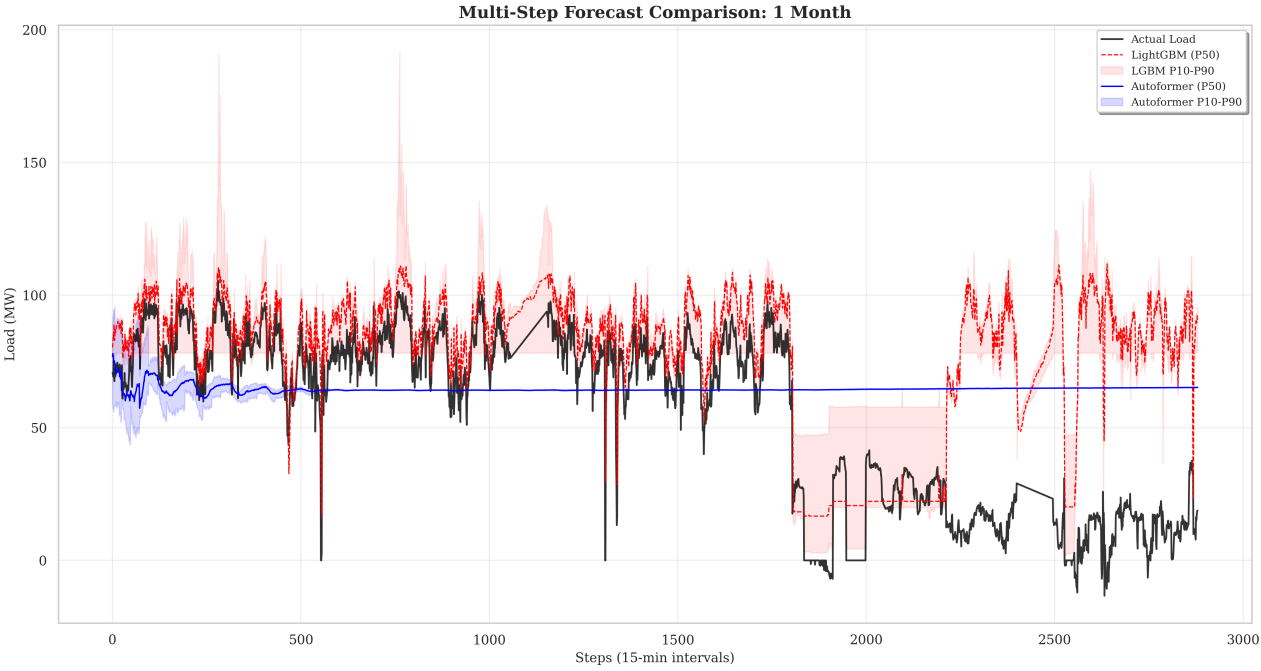


Multi-Step Strategies

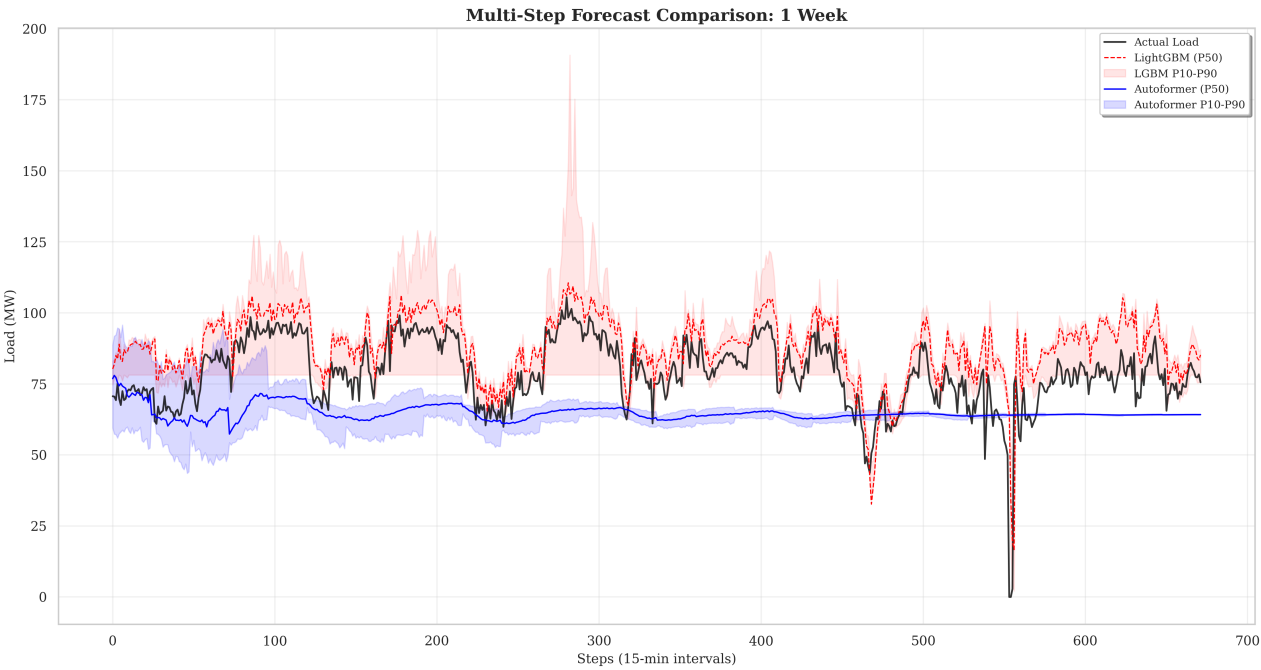
- multi_step/comparison_1_Day.png



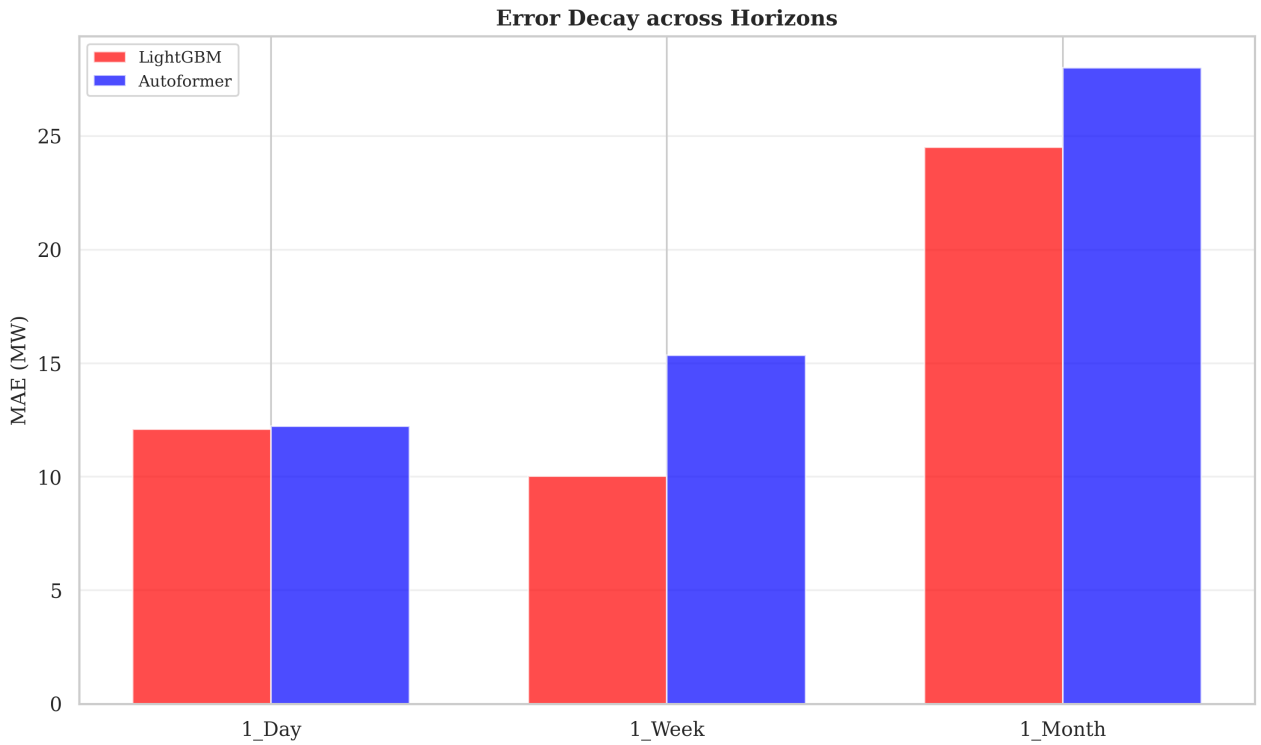
- multi_step/comparison_1_Month.png



- multi_step/comparison_1_Week.png

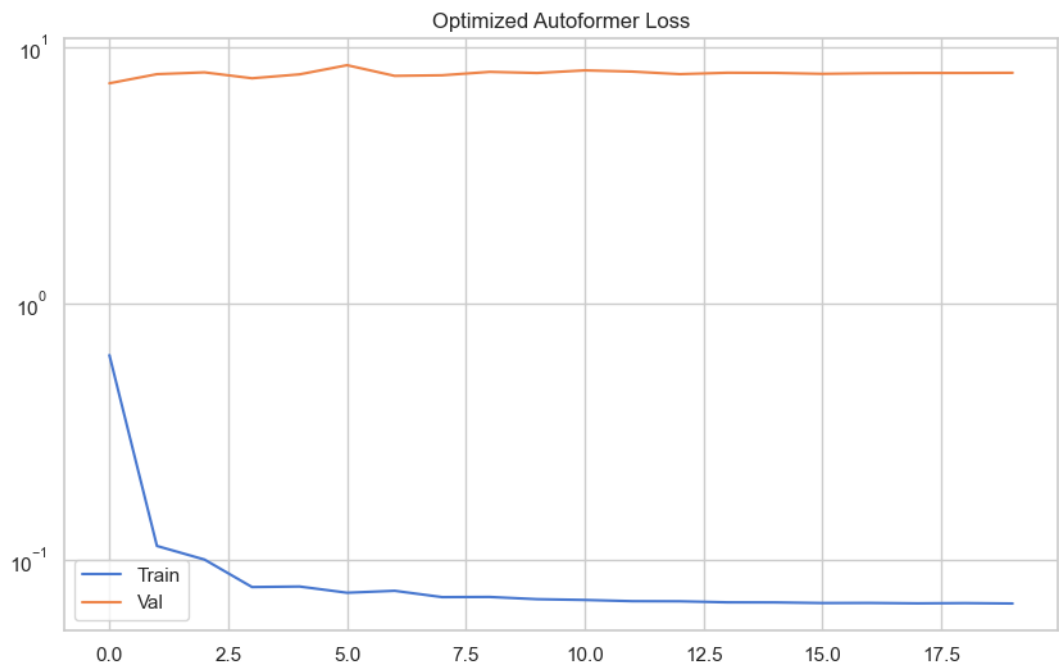


- multi_step/error_decay_comparison.png



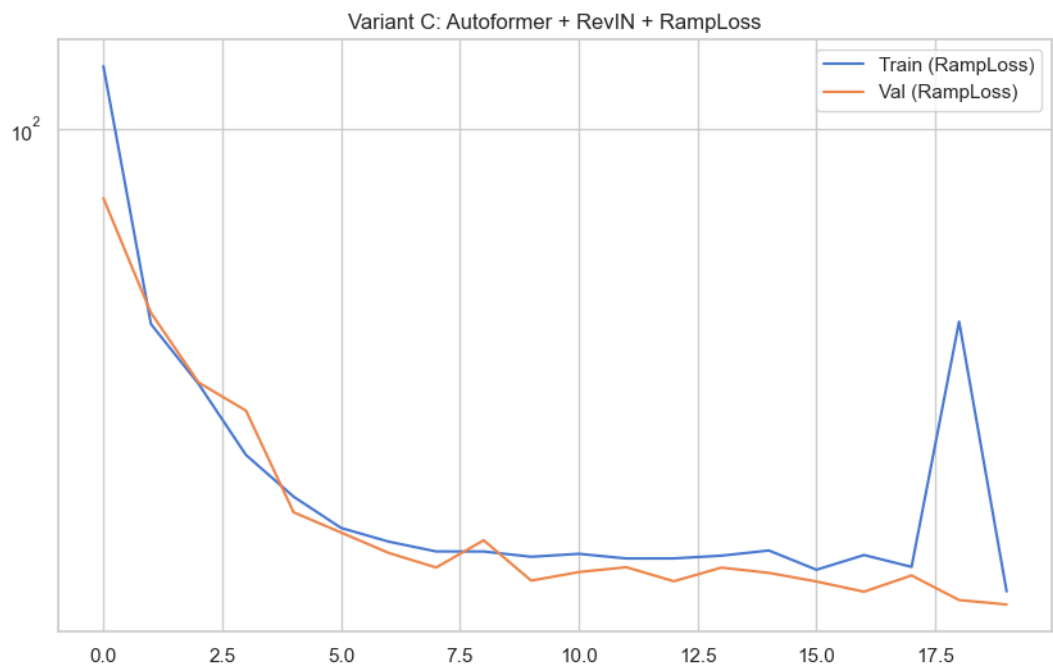
Optimized Autoformer

- **optimized/loss_curve_optimized.png**

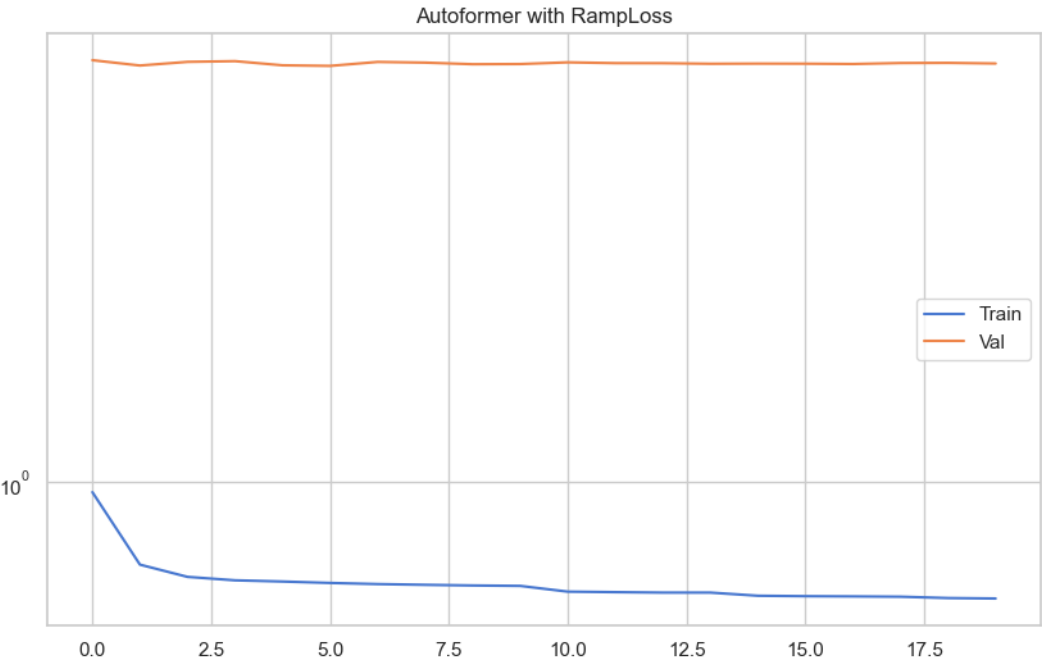


Optimized Ramp Loss

- **optimized_ramp/loss_curve_final.png**

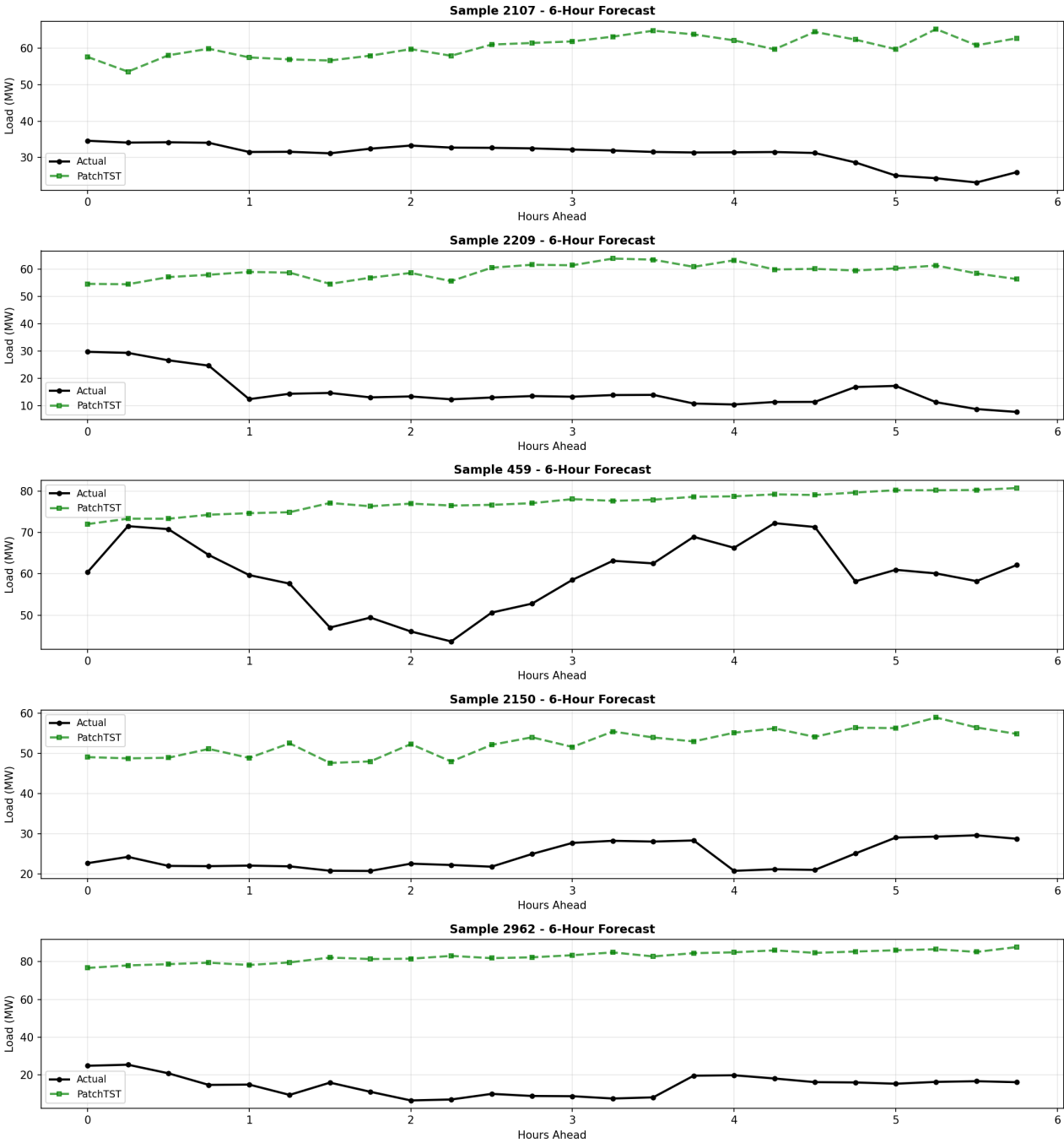


- **optimized_ramp/loss_curve_ramp.png**

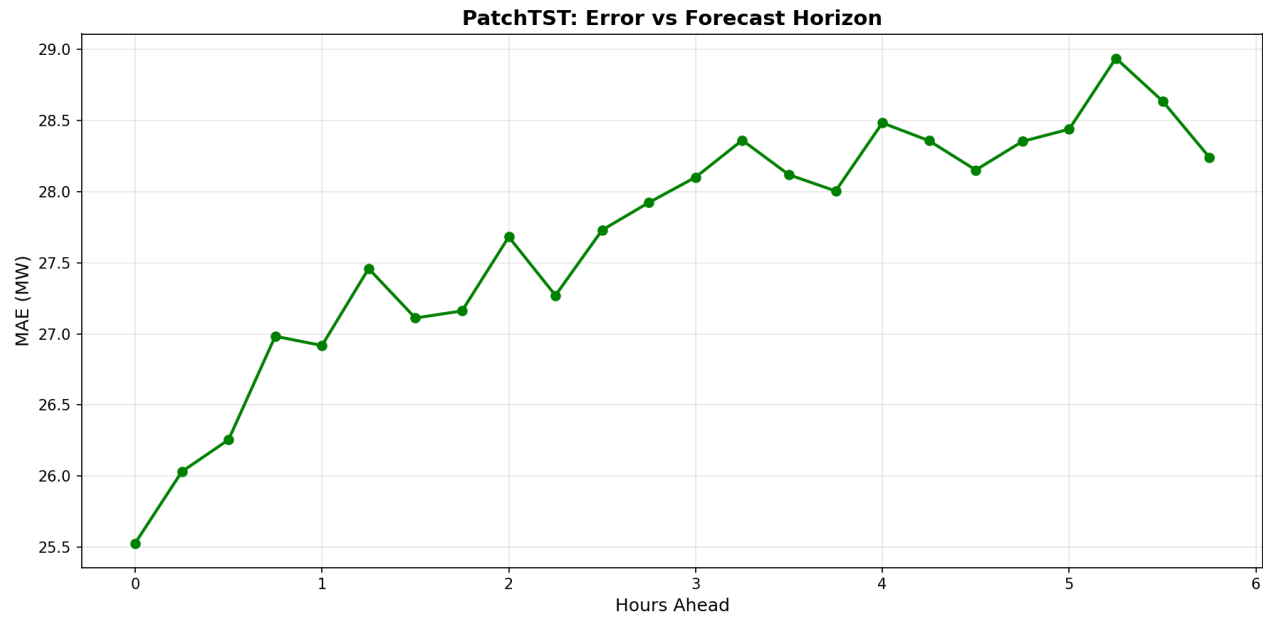


PatchTST

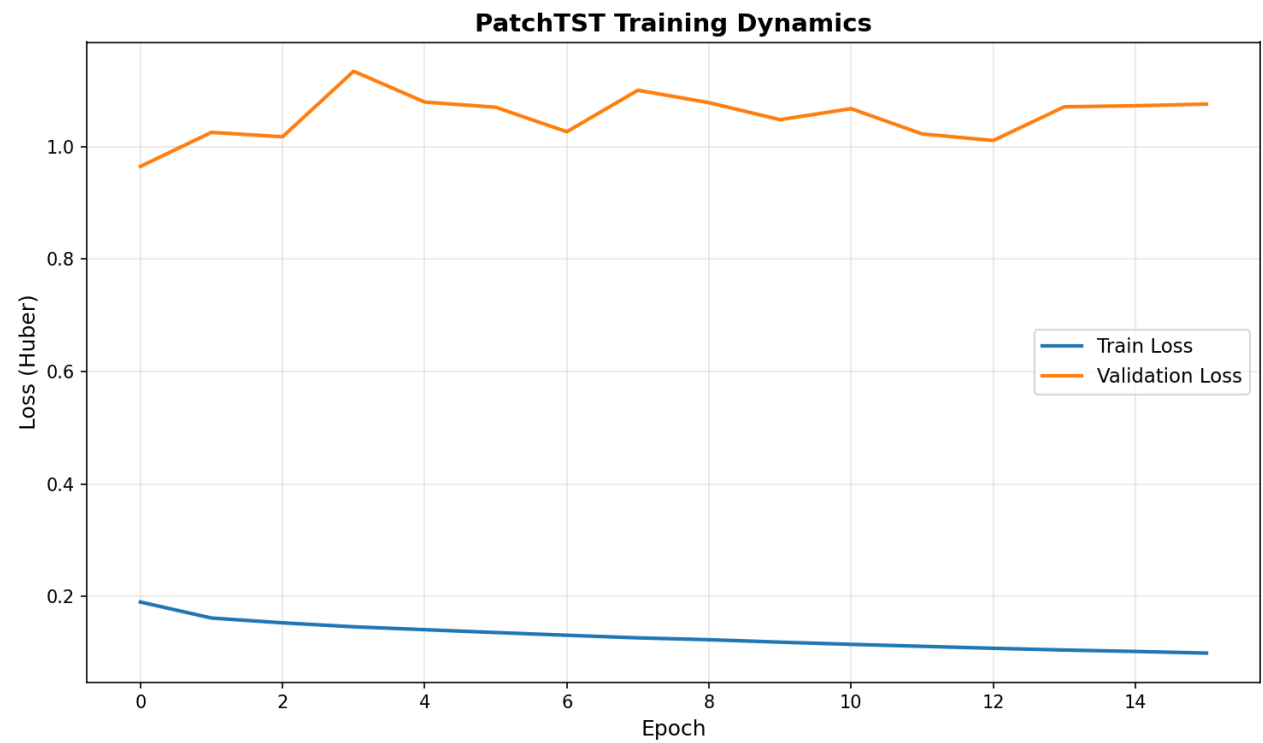
• patchtst/forecast_samples.png



- patchtst/horizon_mae.png

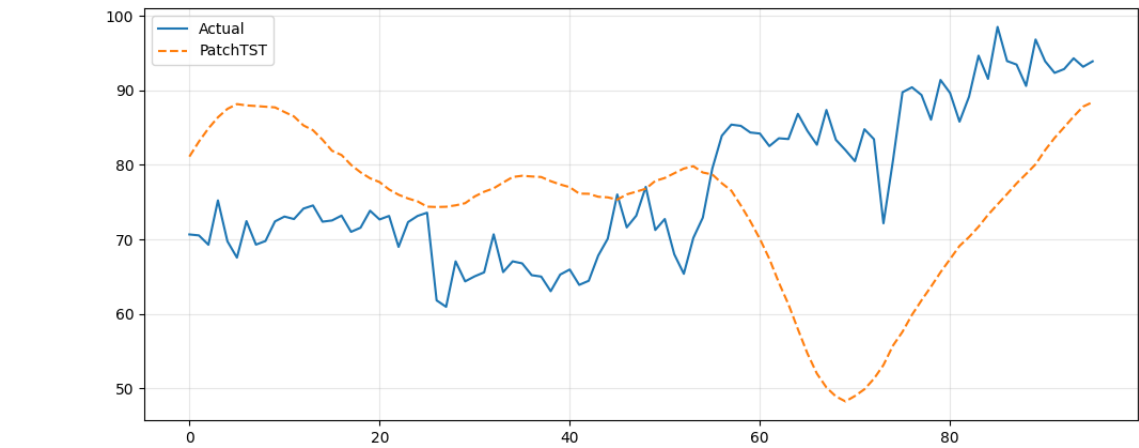


- patchtst/training_curves.png



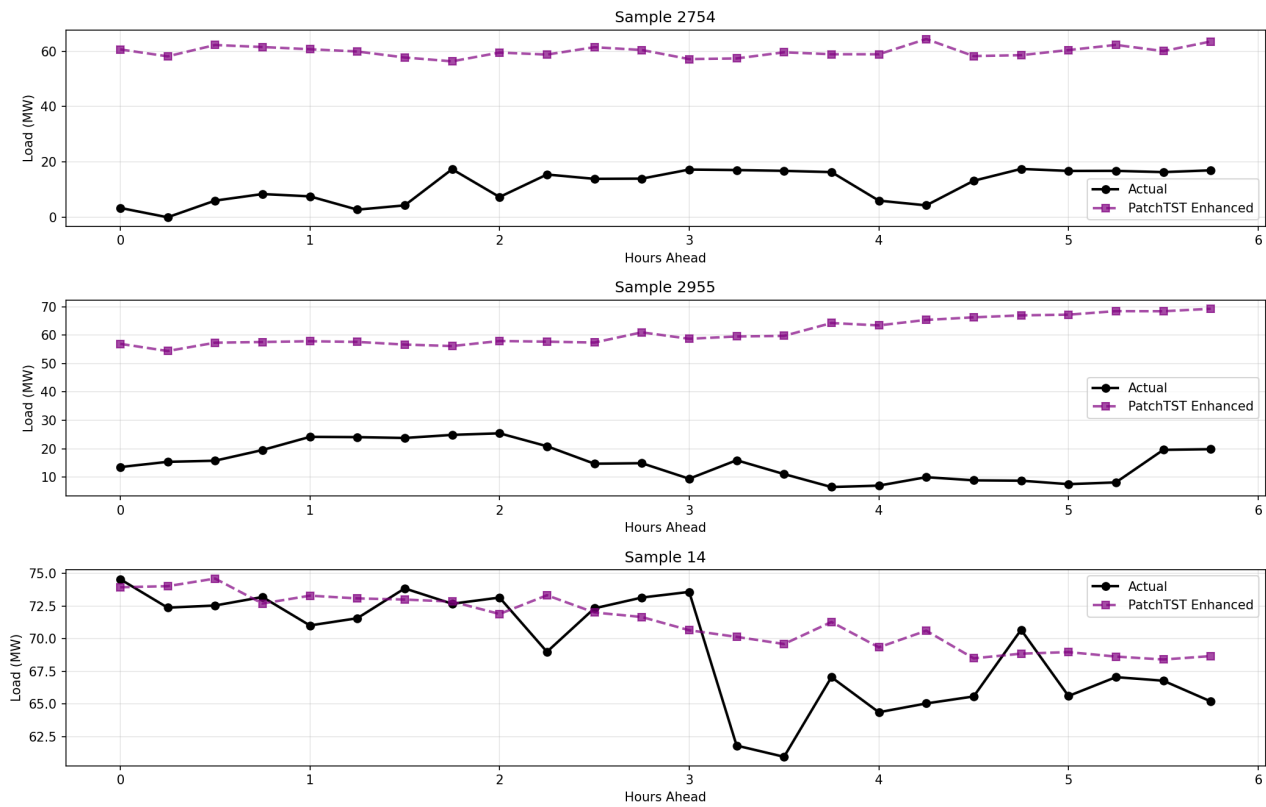
PatchTST Clean

- patchtst_clean/forecast.png

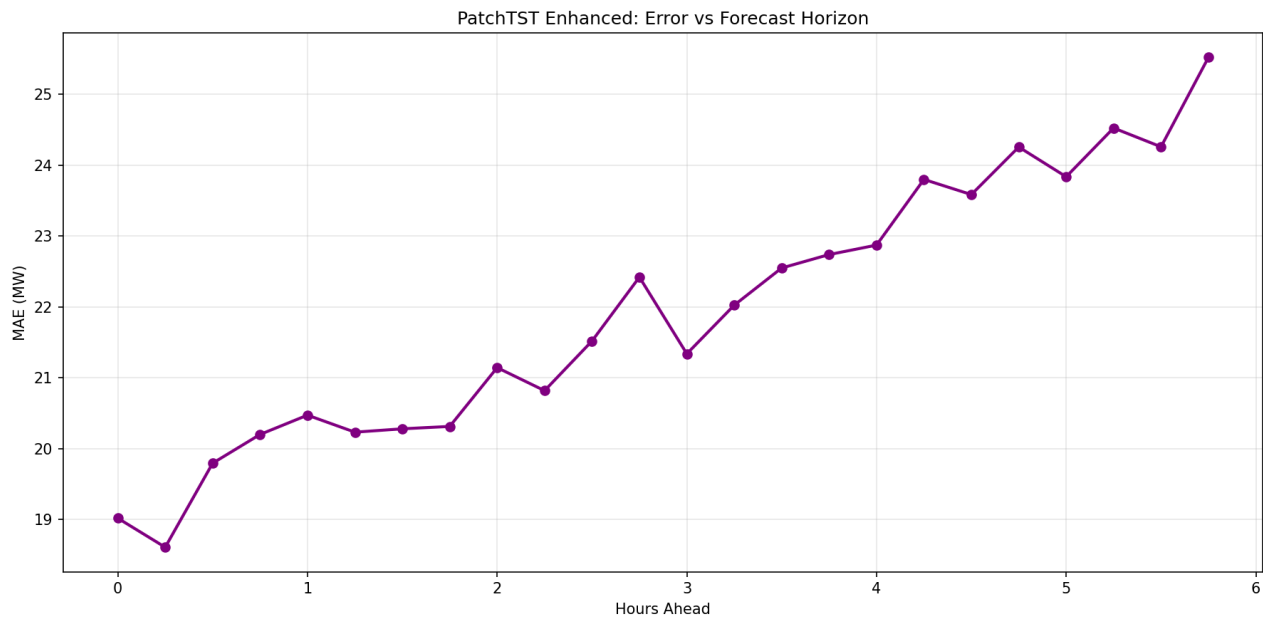


PatchTST Enhanced

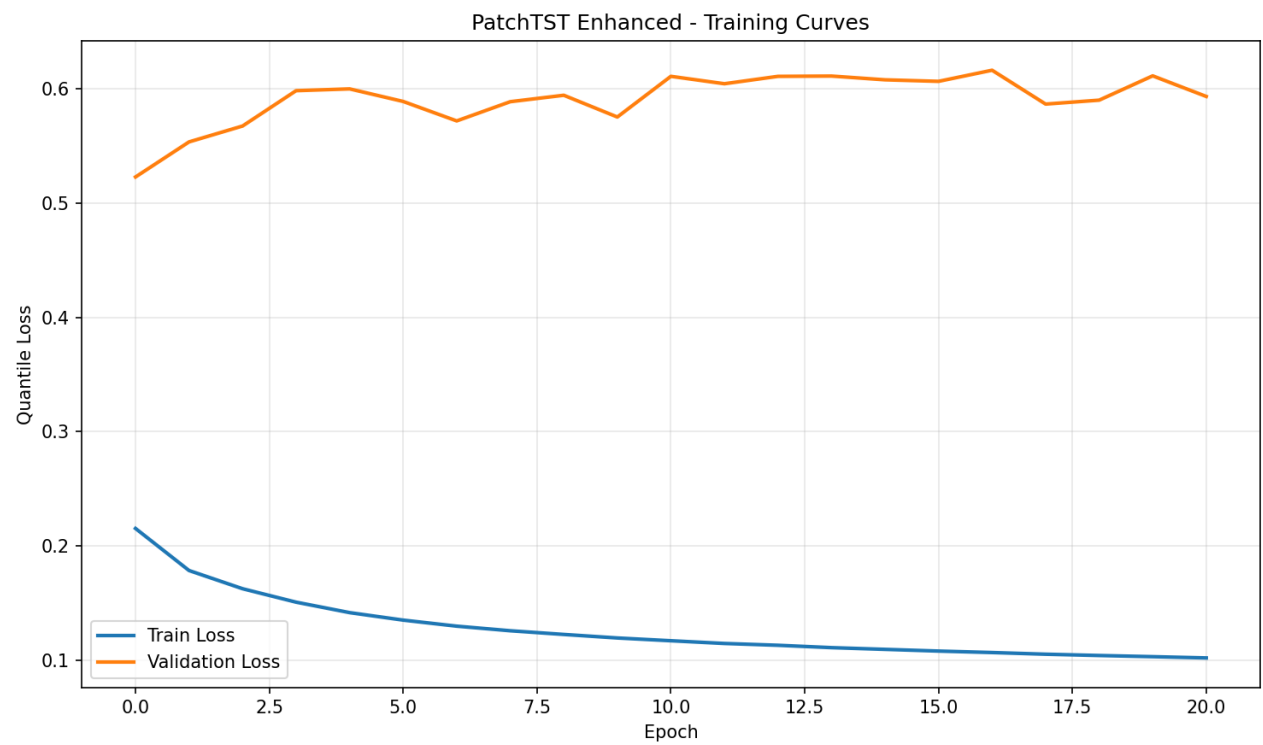
- patchtst_enhanced/forecast_samples.png



- patchtst_enhanced/horizon_mae.png

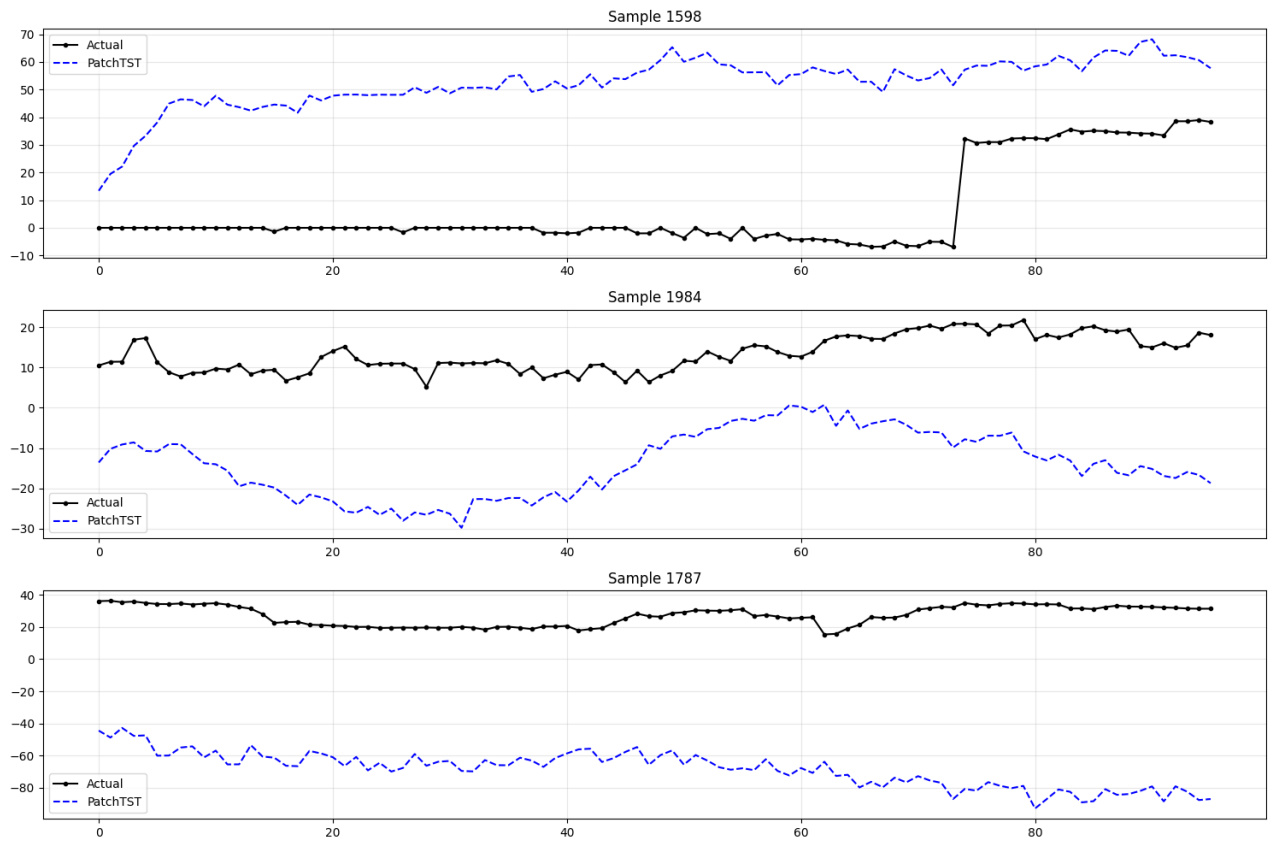


- patchtst_enhanced/training_curves.png

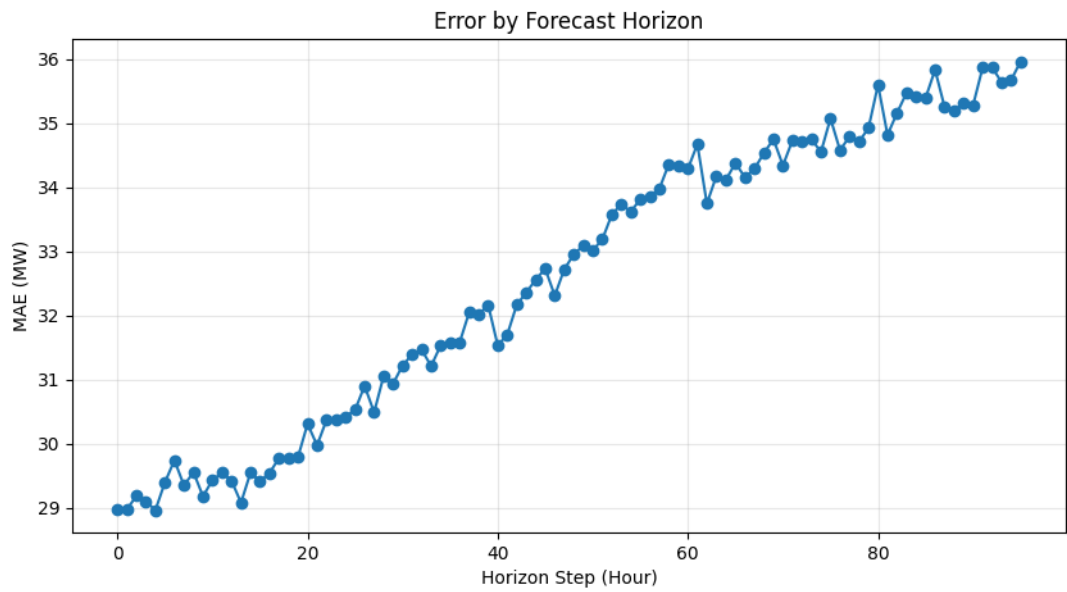


PatchTST Refactored

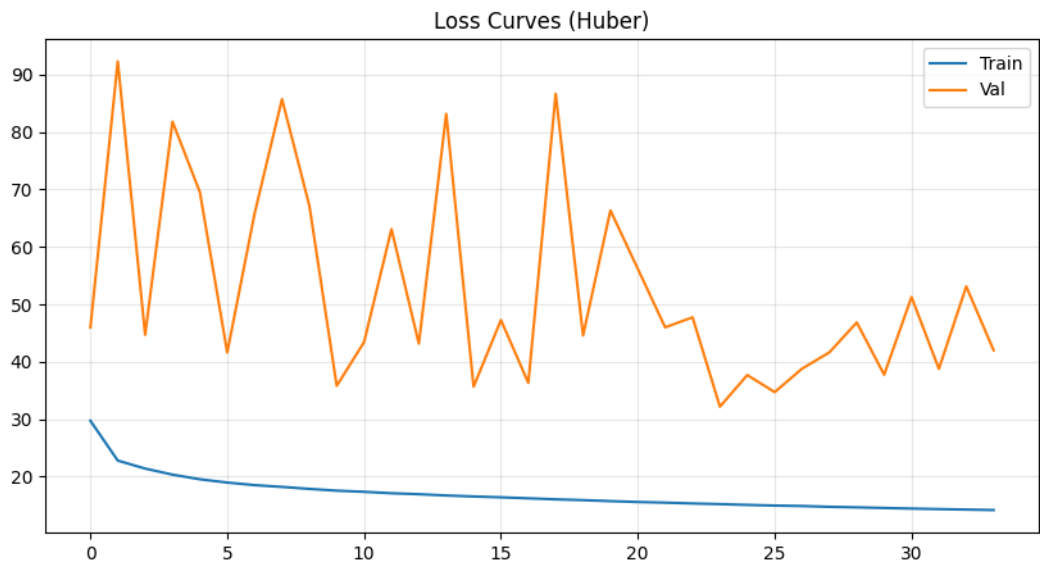
- patchtst_refactored/forecast_examples.png



- patchtst_refactored/horizon_error.png

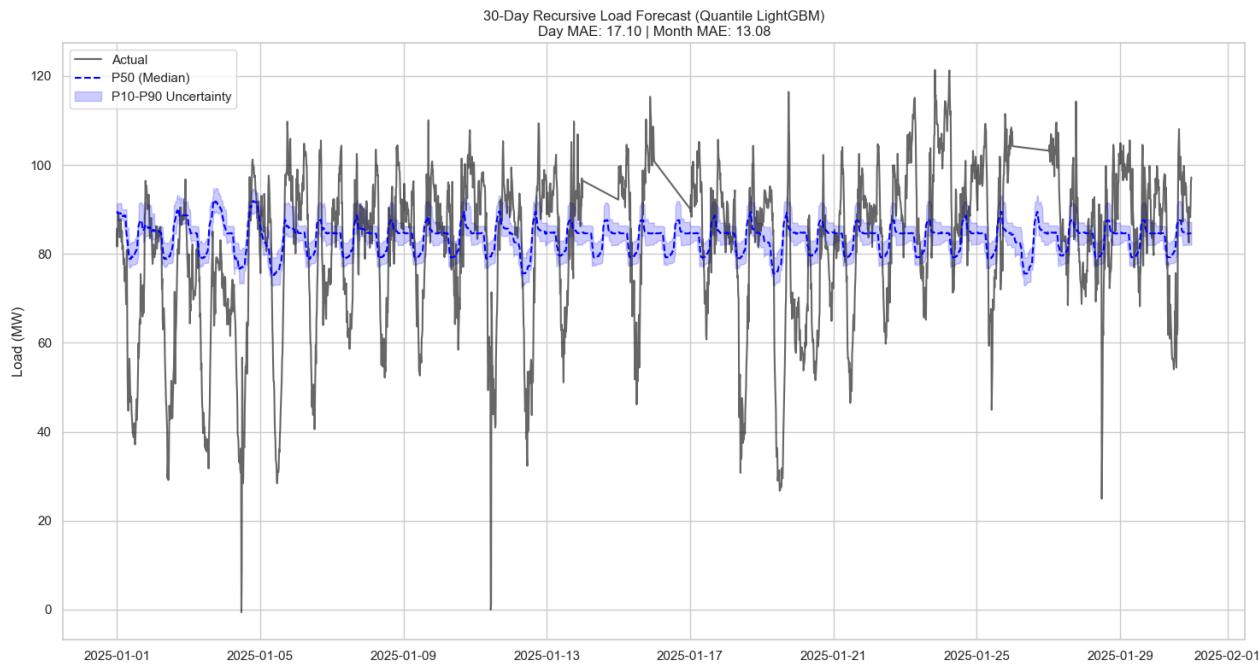


- patchtst_refactored/loss_curve.png



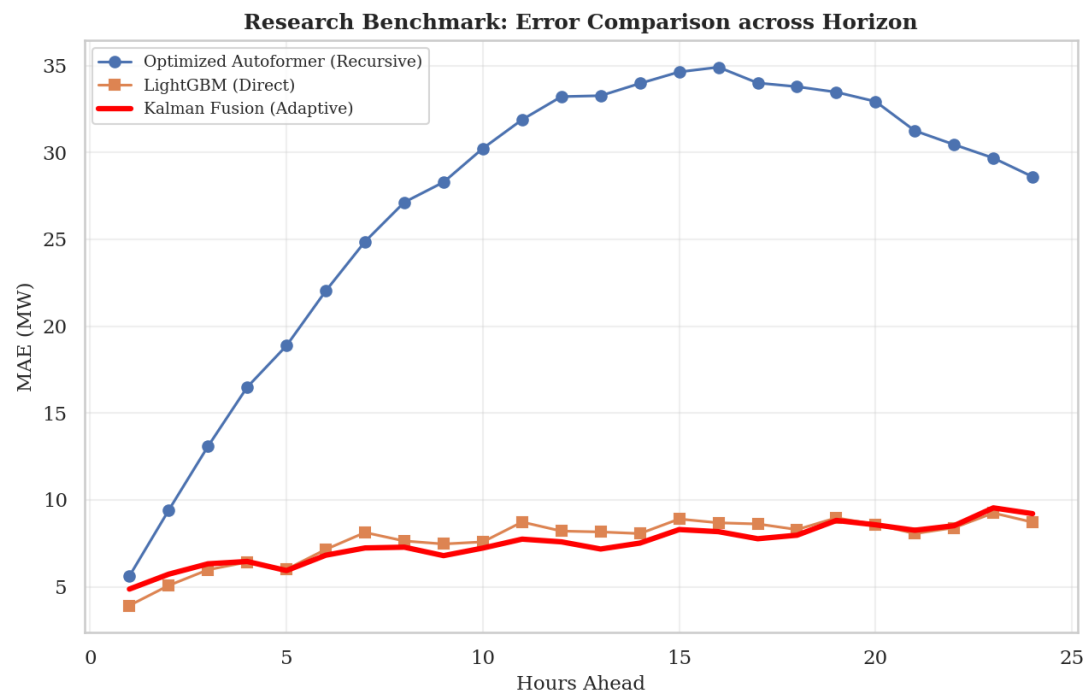
Recursive Quantile

- recursive_quantile/recursive_month_quantile.png

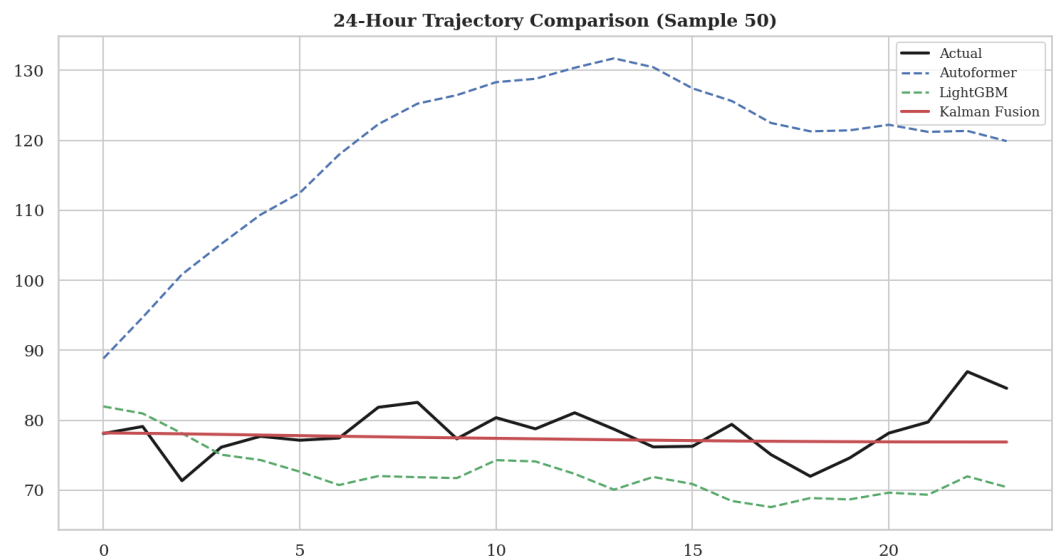


Research Benchmarks

- research_benchmark/benchmark_mae_profile.png

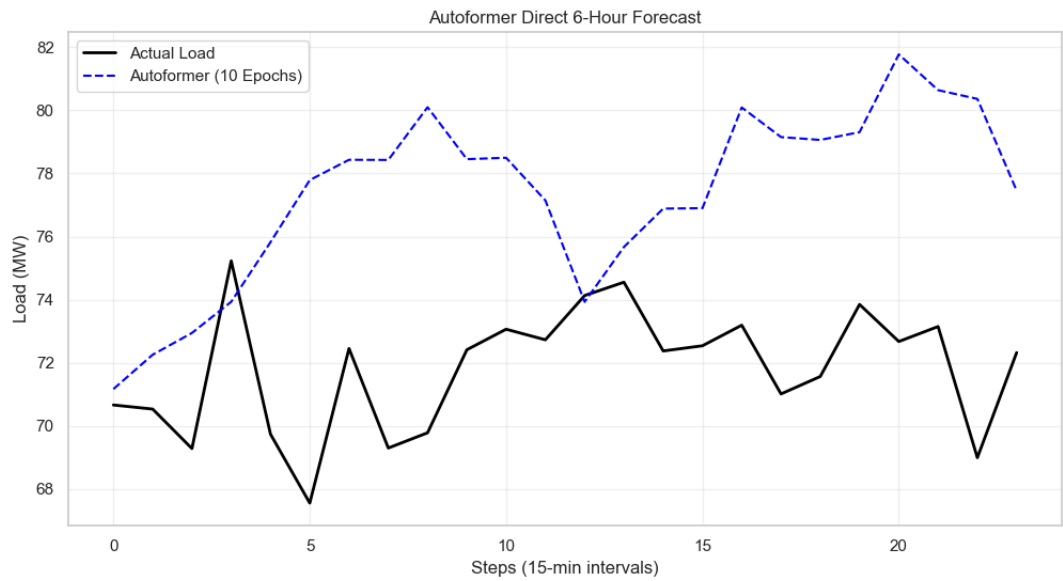


- research_benchmark/benchmark_trajectory.png

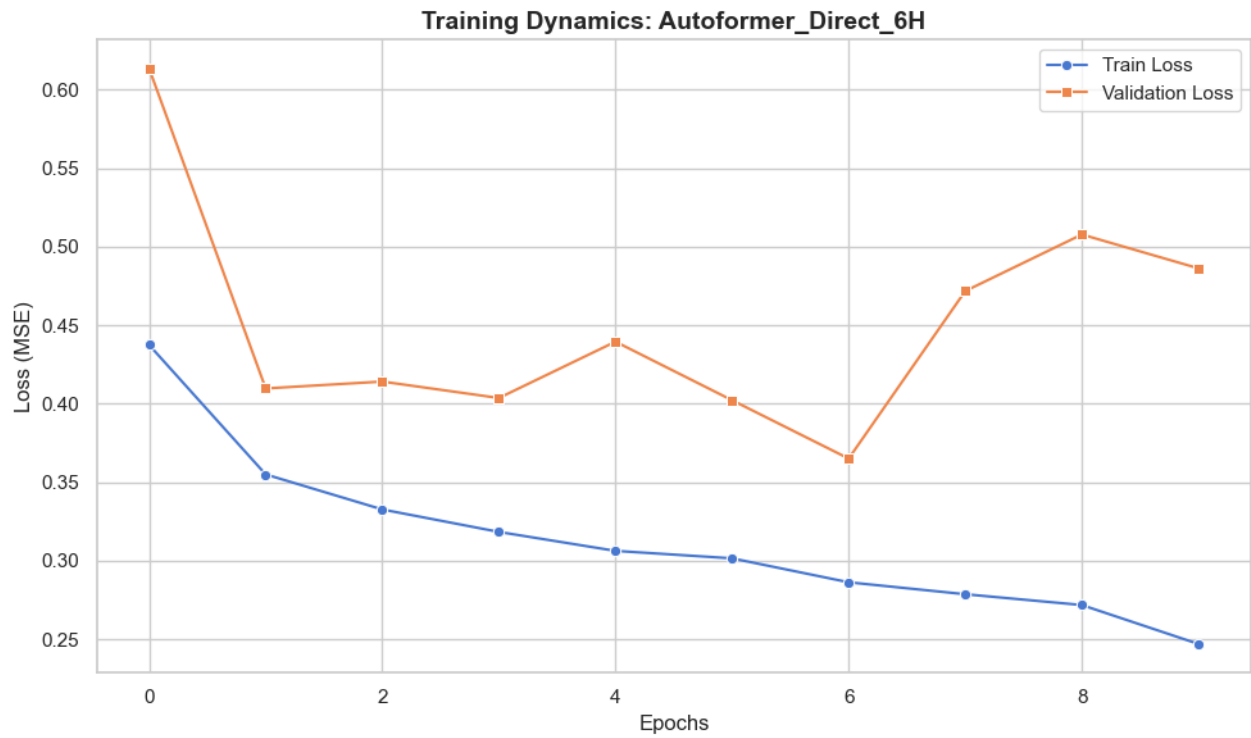


SOTA (Autoformer/Informer)

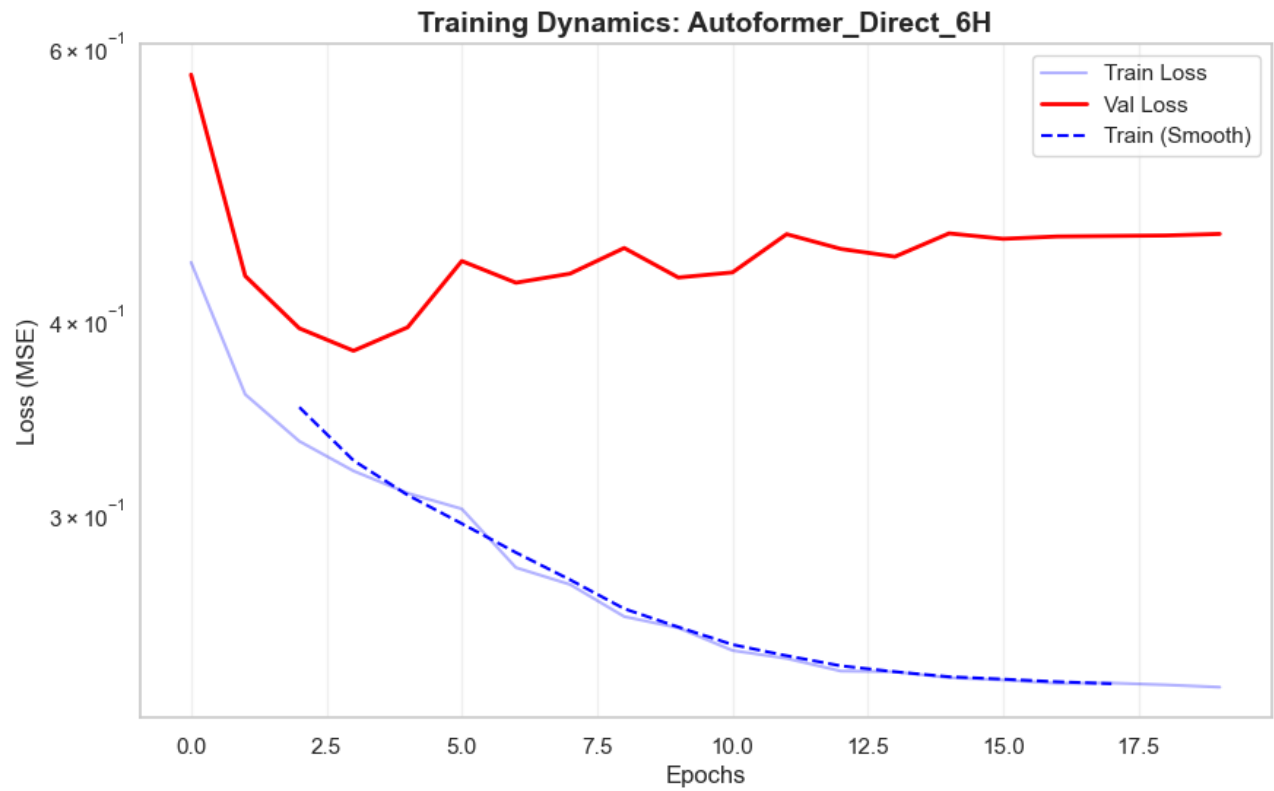
- **sota/Autoformer_Direct_6H_Forecast.png**



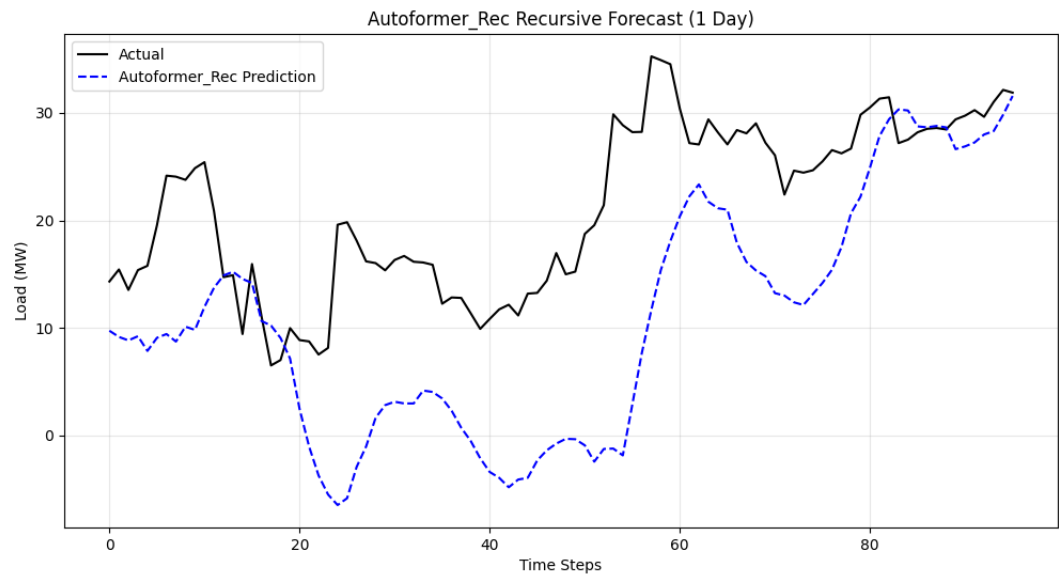
- **sota/Autoformer_Direct_6H_loss.png**



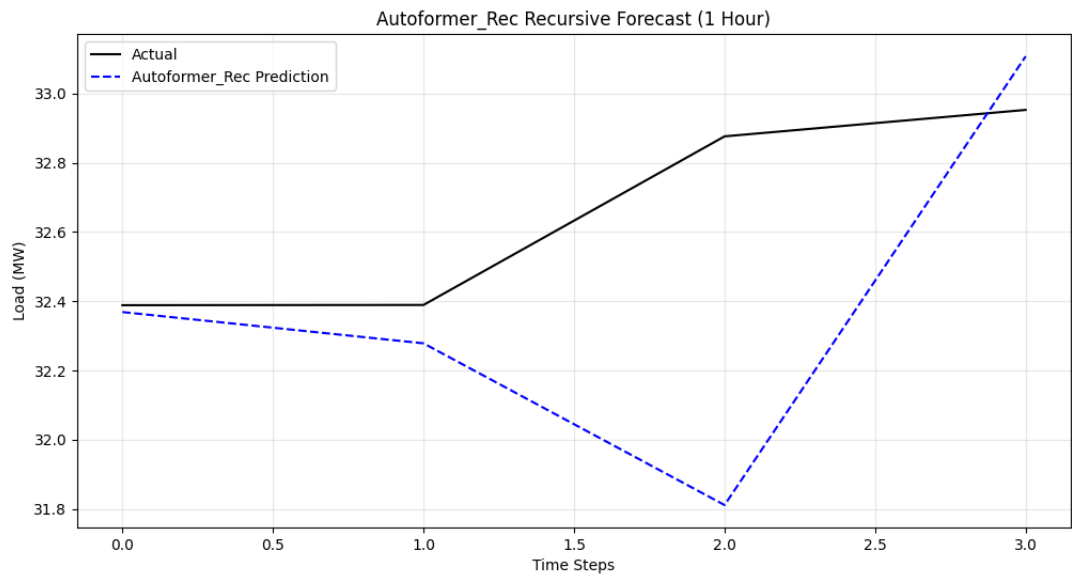
- **sota/Autoformer_Direct_6H_loss_curve.png**



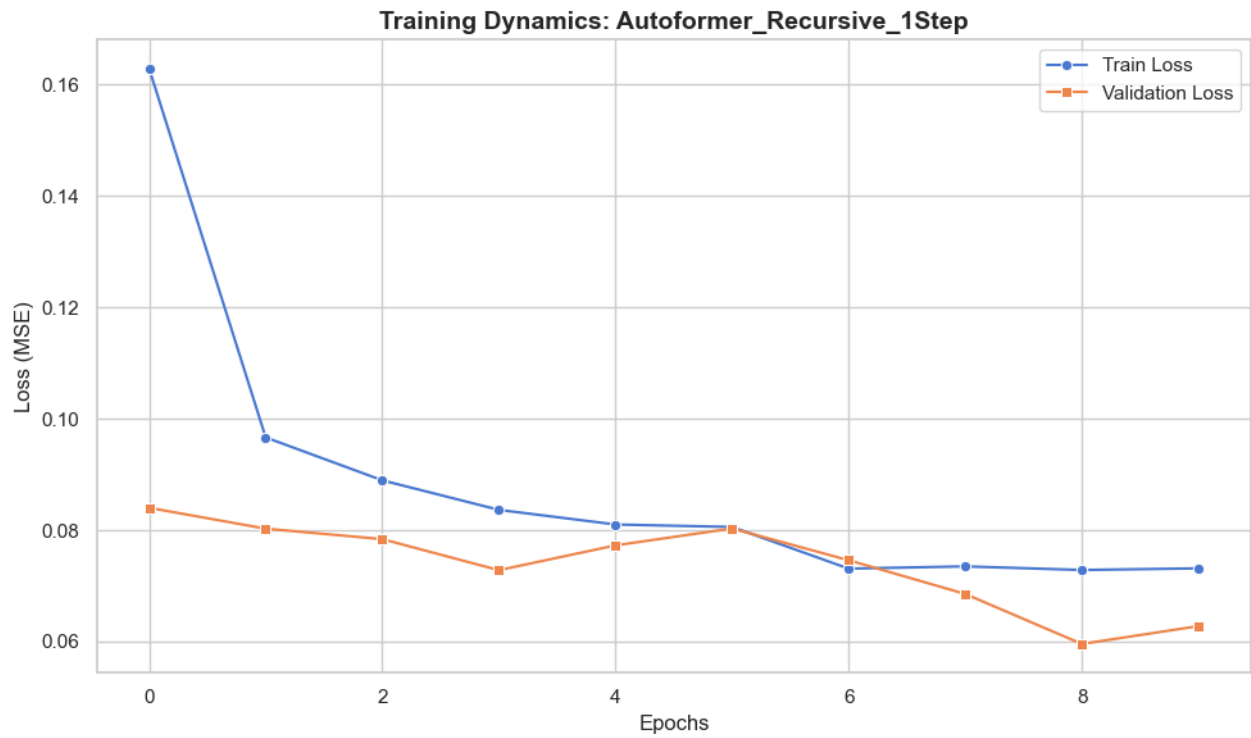
- **sota/Autoformer_Rec_recursive_1_Day.png**



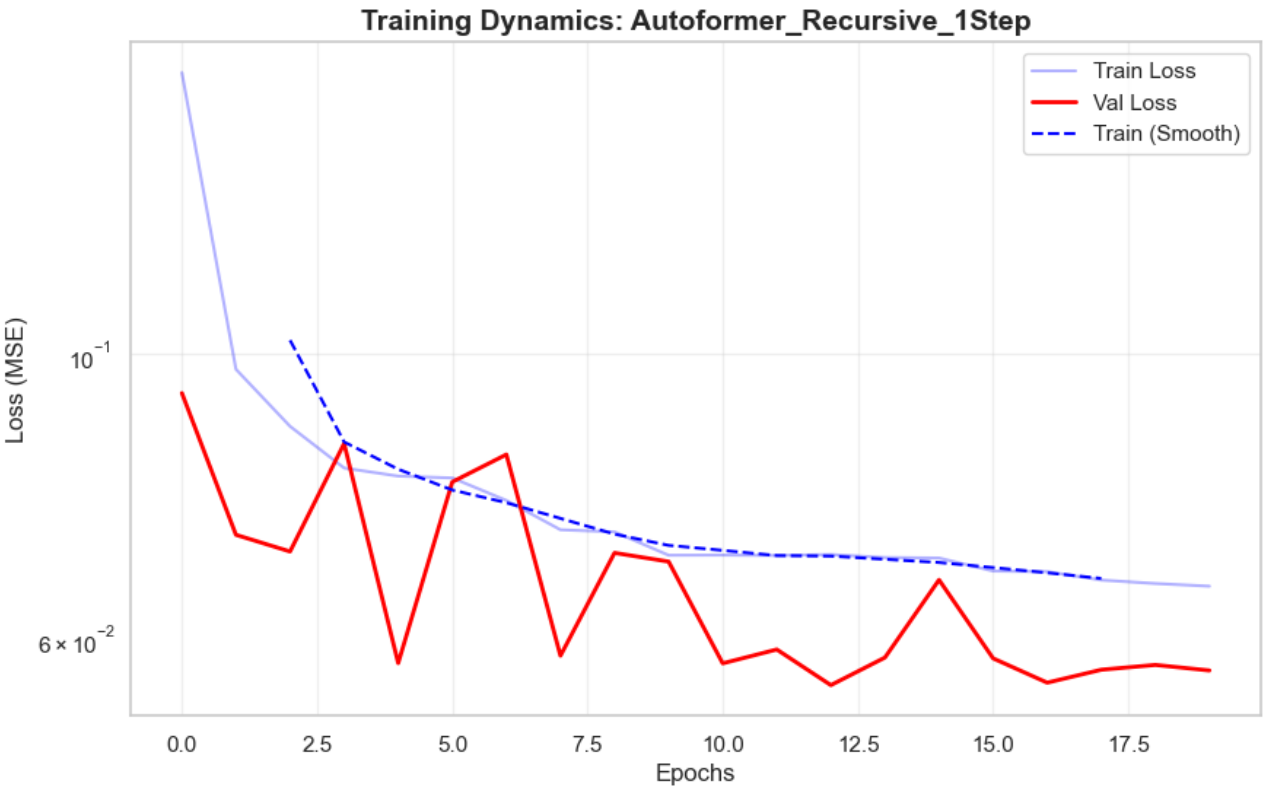
- **sota/Autoformer_Rec_recursive_1_Hour.png**



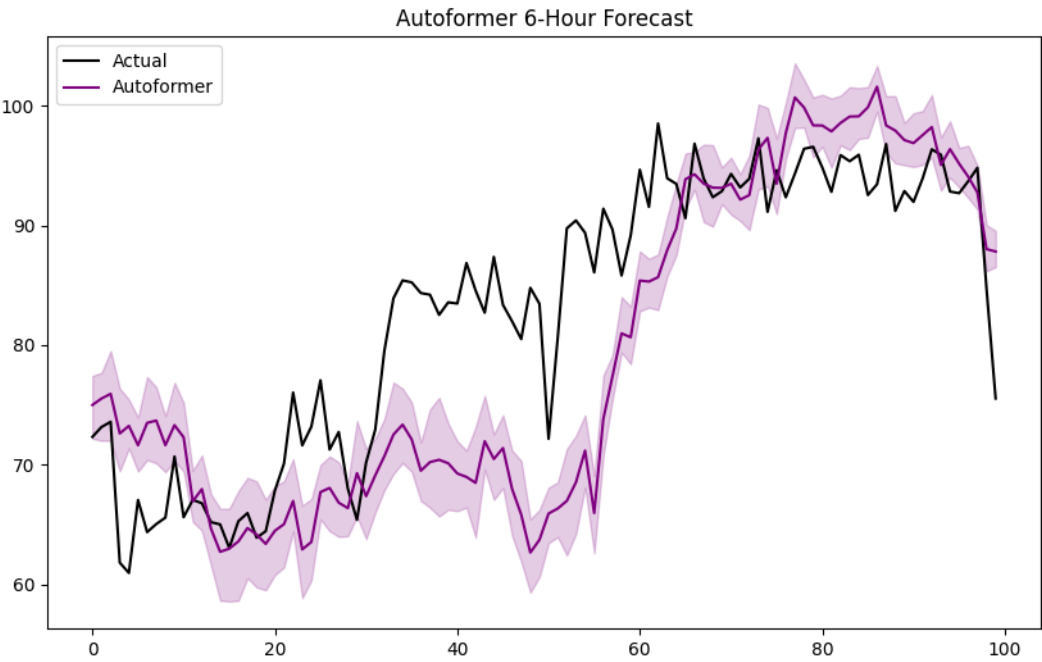
- **sota/Autoformer_Recursive_1Step_loss.png**



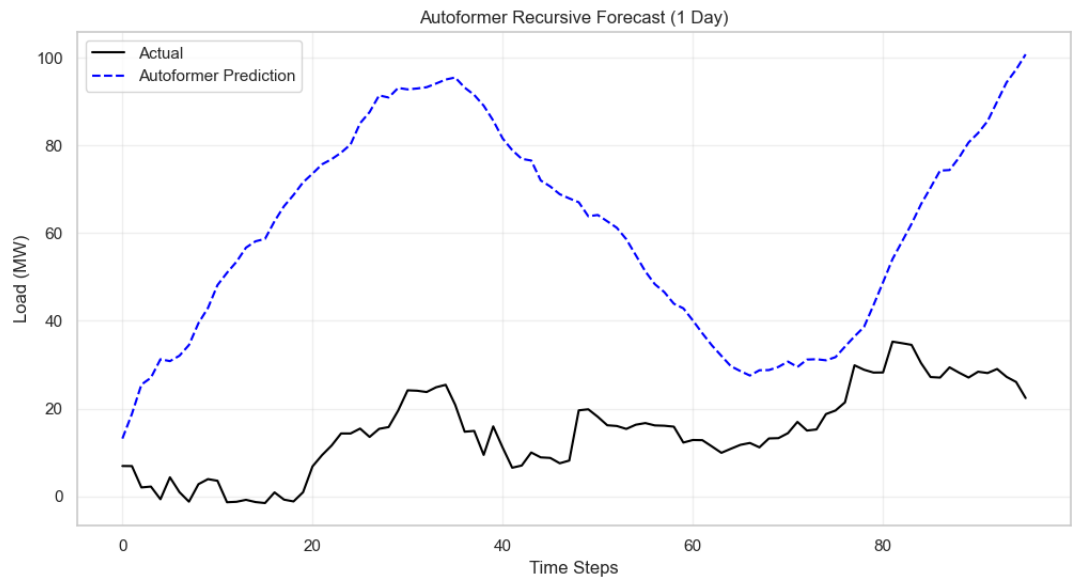
- **sota/Autoformer_Recursive_1Step_loss_curve.png**



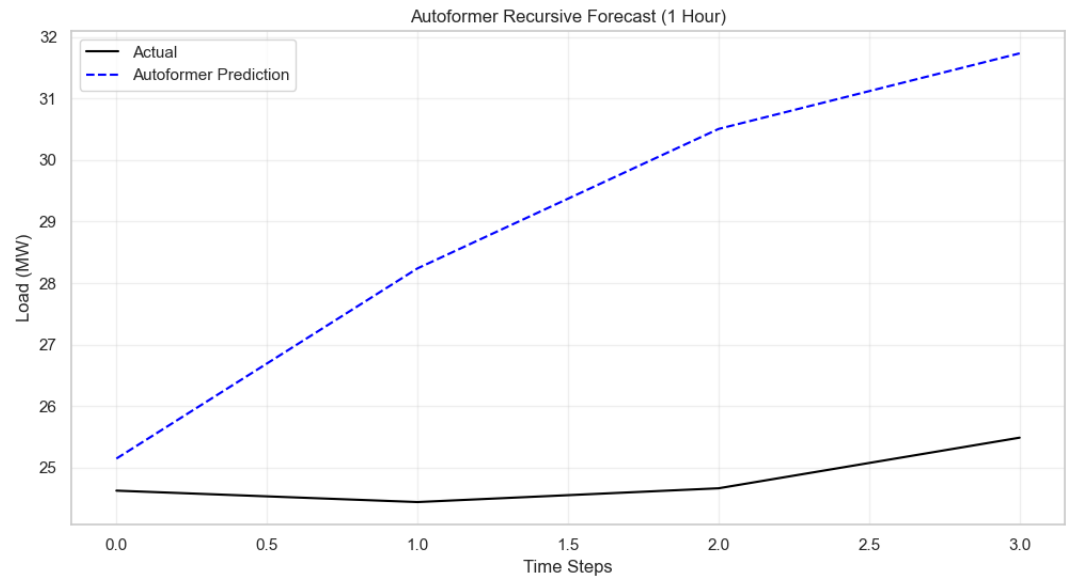
- **sota/Autoformer_forecast.png**



- **sota/Autoformer_recursive_1_Day.png**

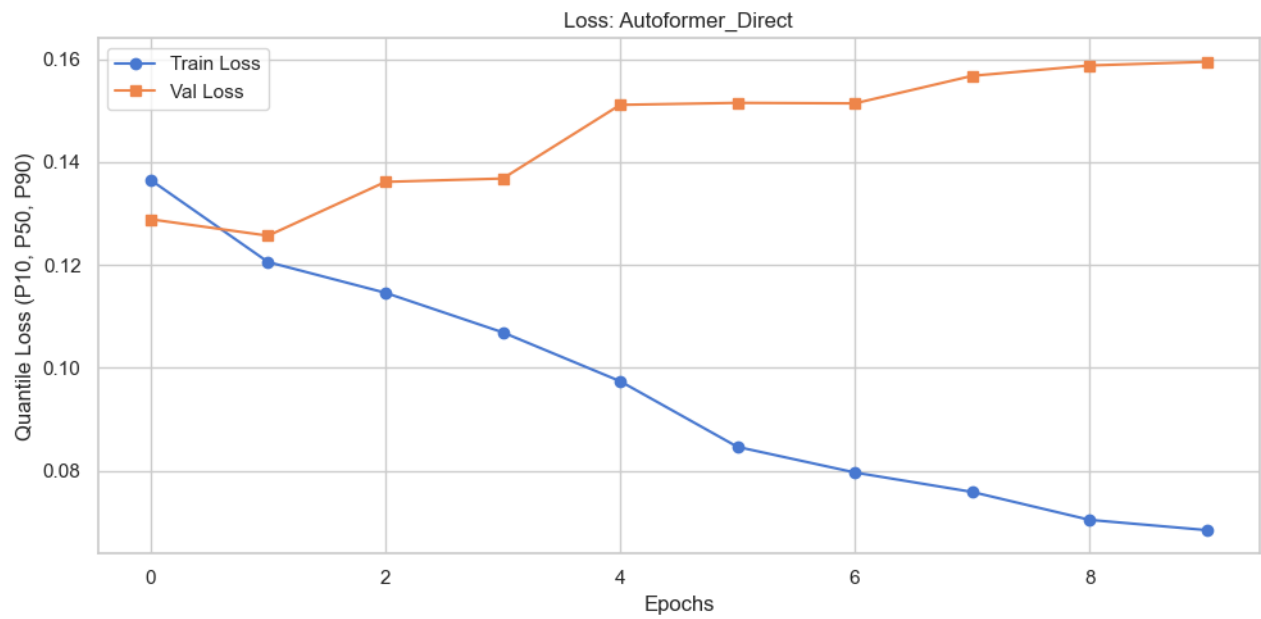


- **sota/Autoformer_recursive_1_Hour.png**

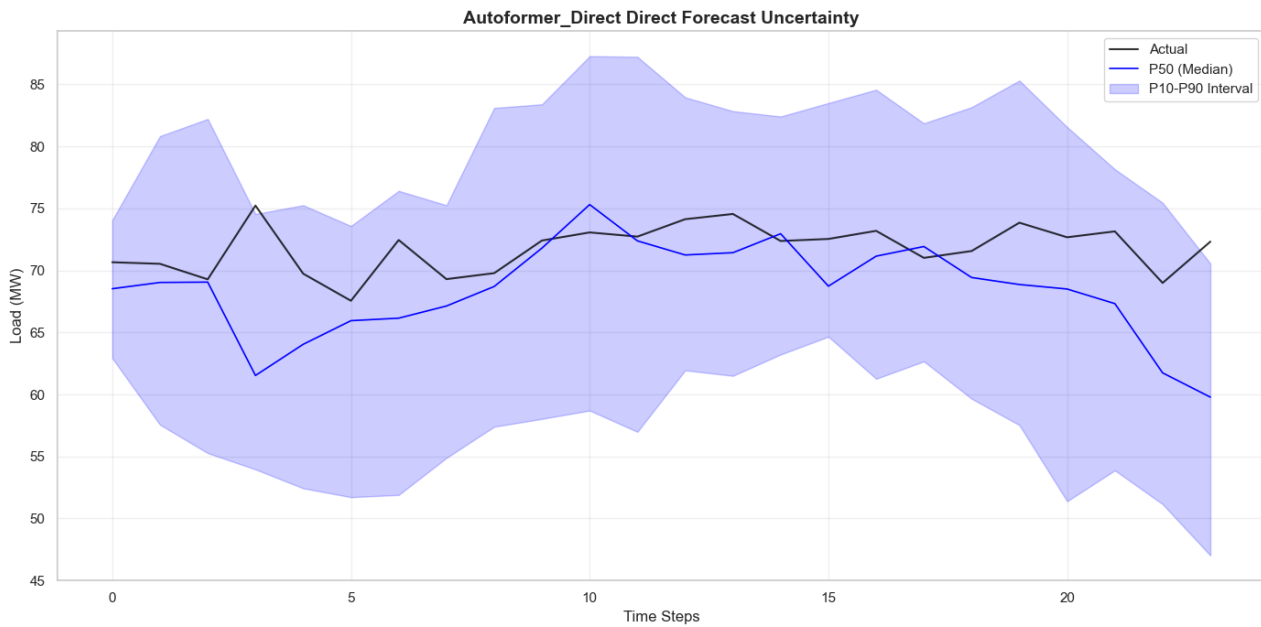


SOTA Robust (RevIN Fixed)

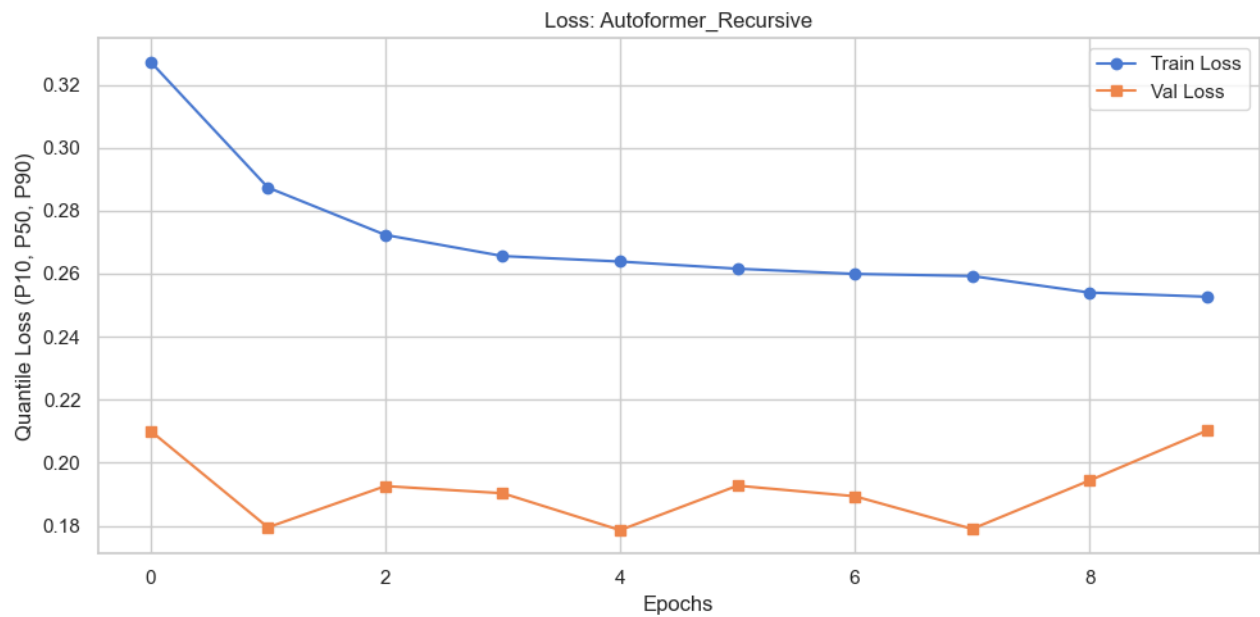
- **sota_robust/Autoformer_Direct_loss.png**



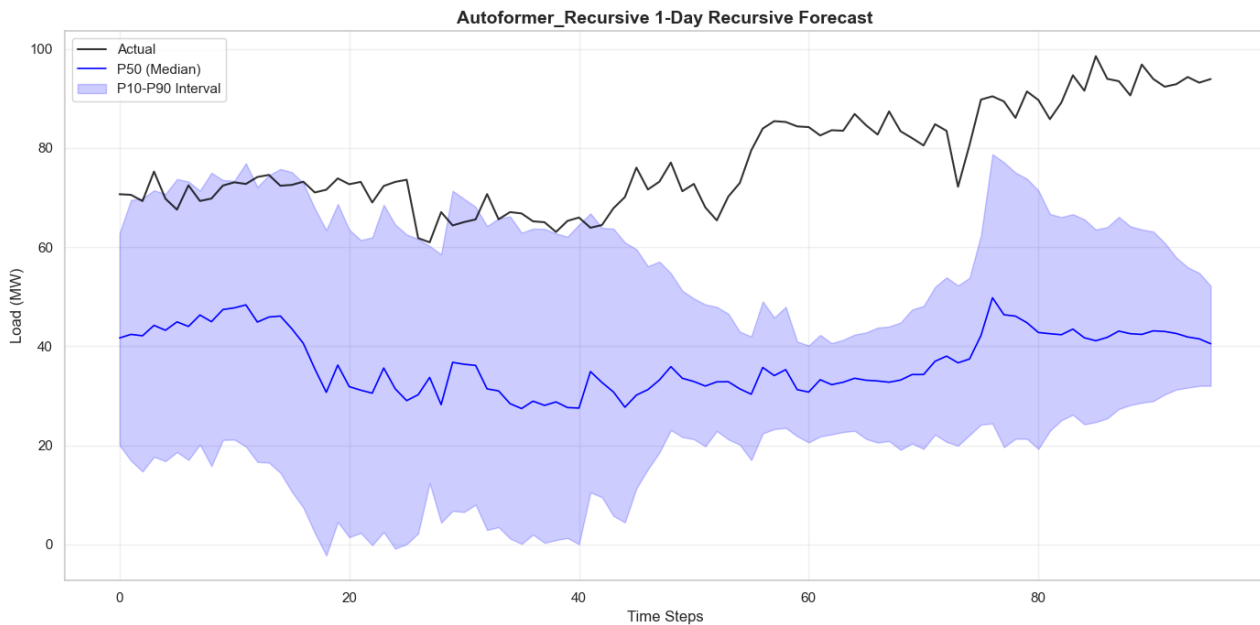
- **sota_robust/Autoformer_Direct_uncertainty.png**



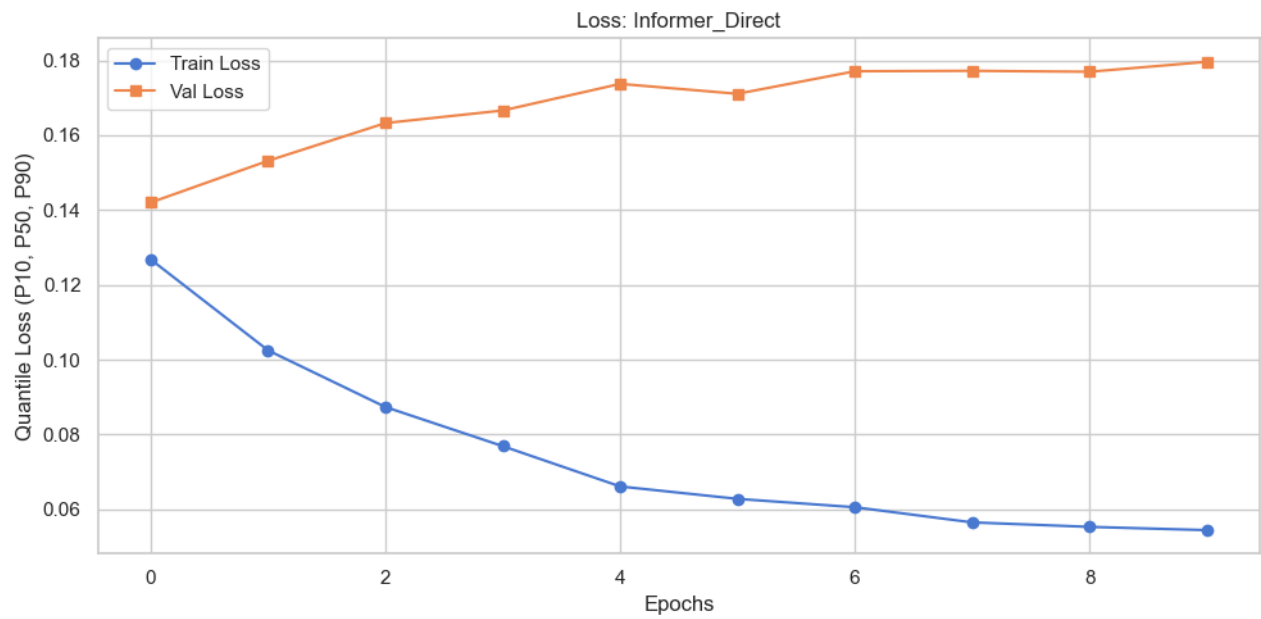
- **sota_robust/Autoformer_Recursive_loss.png**



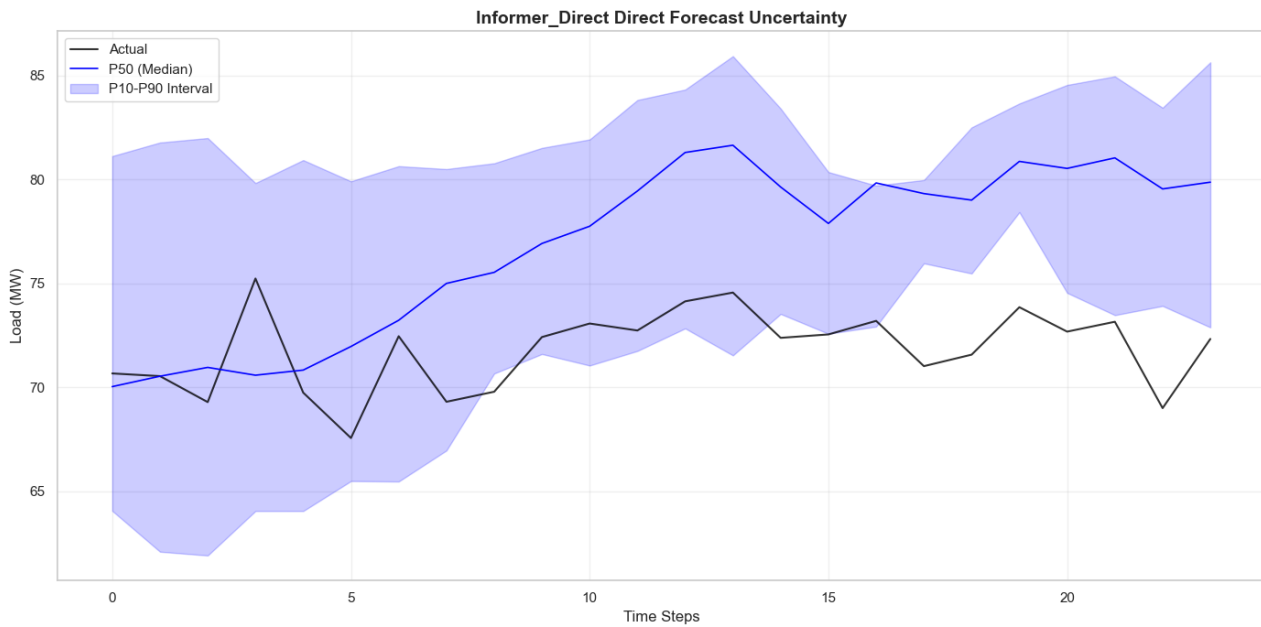
- **sota_robust/Autoformer_Recursive_recursive_uncertainty.png**



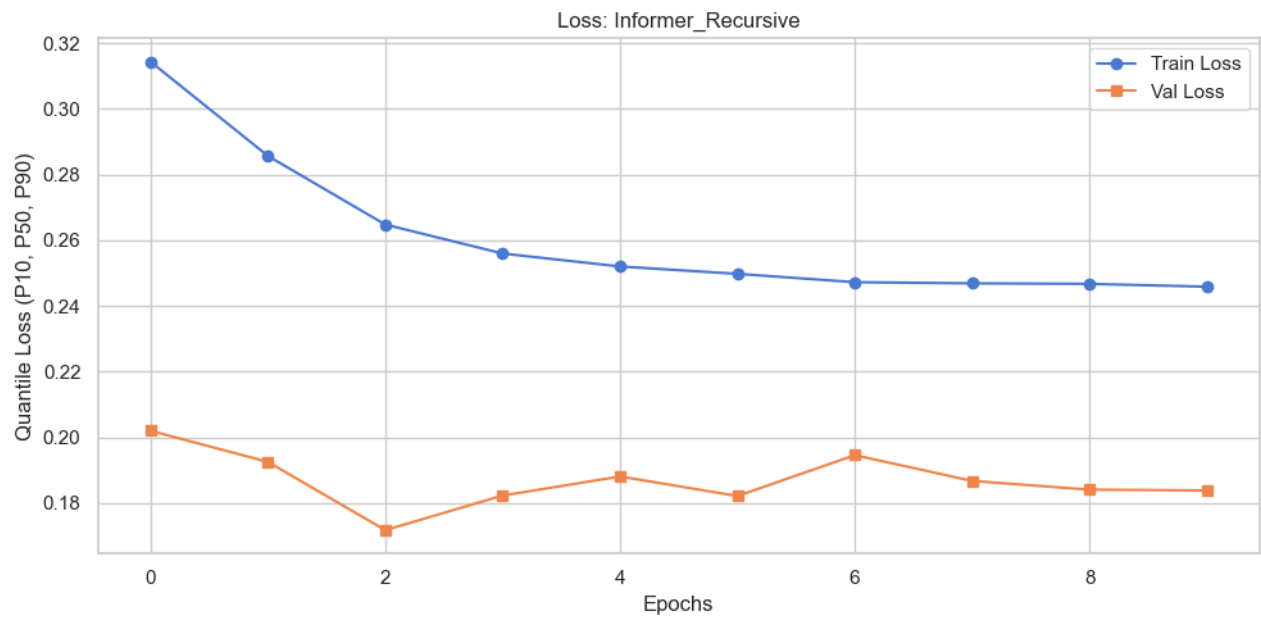
- **sota_robust/Informer_Direct_loss.png**



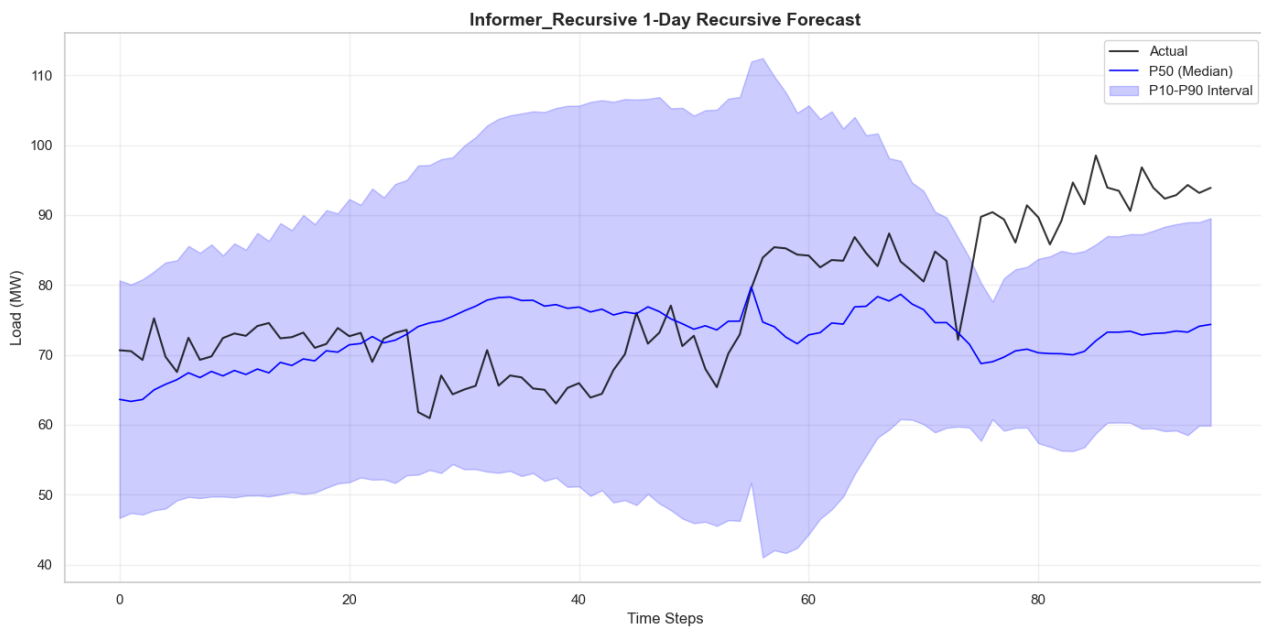
- **sota_robust/Informer_Direct_uncertainty.png**



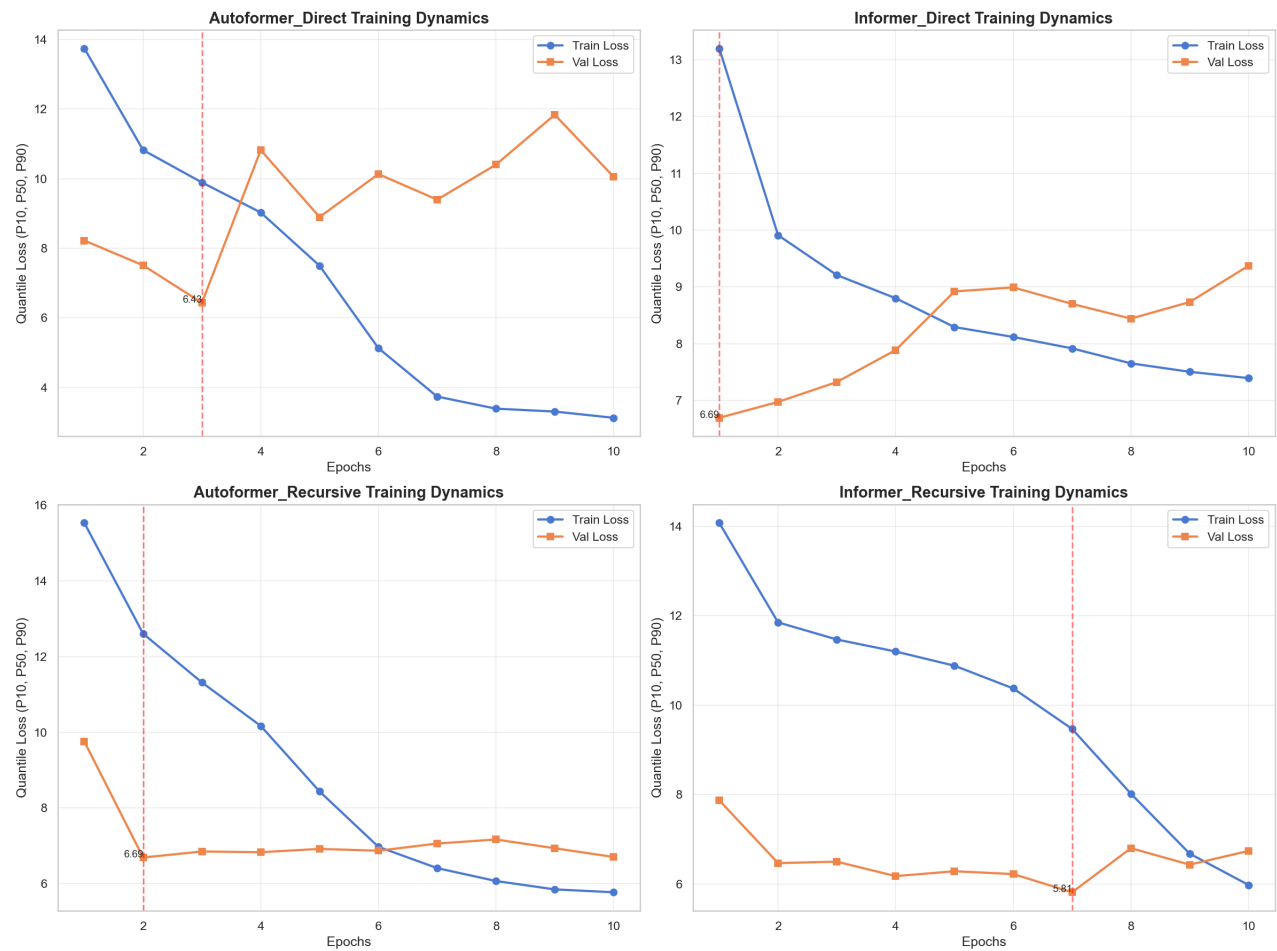
- **sota_robust/Informer_Recursive_loss.png**



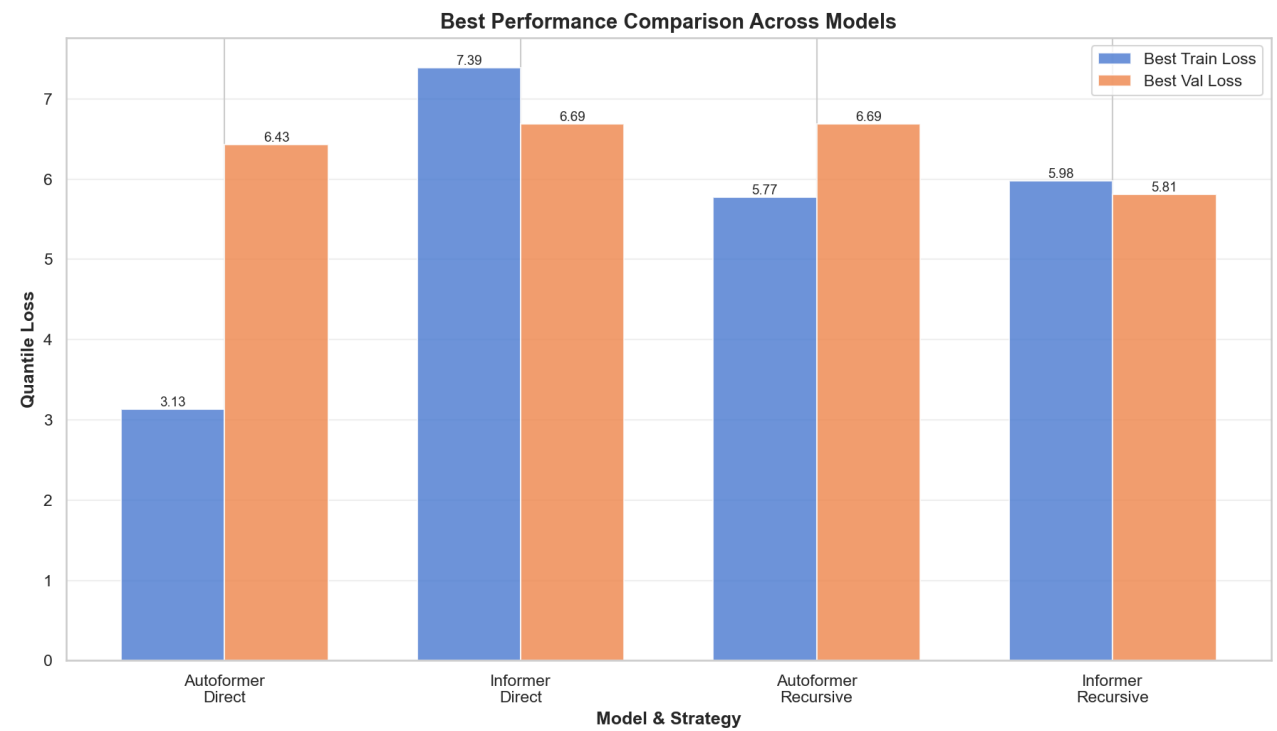
- **sota_robust/Informer_Recursive_recursive_uncertainty.png**



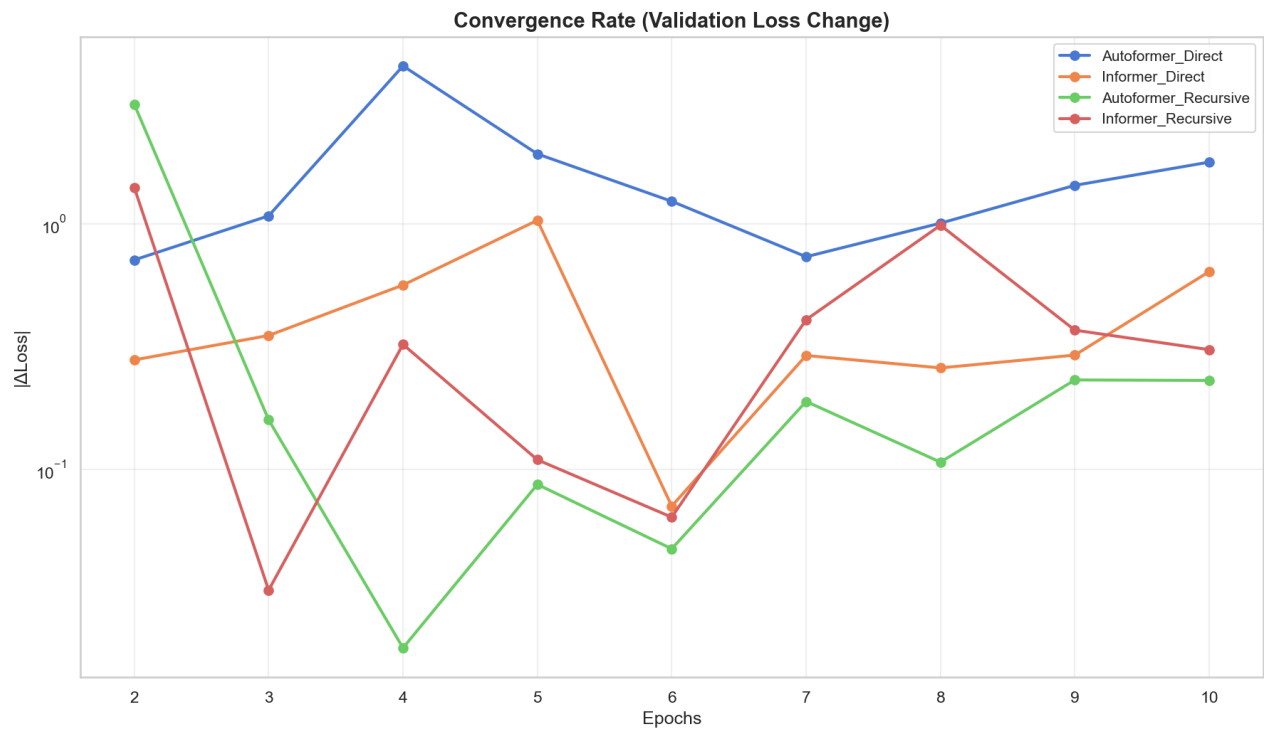
• **sota_robust/all_models_training.png**



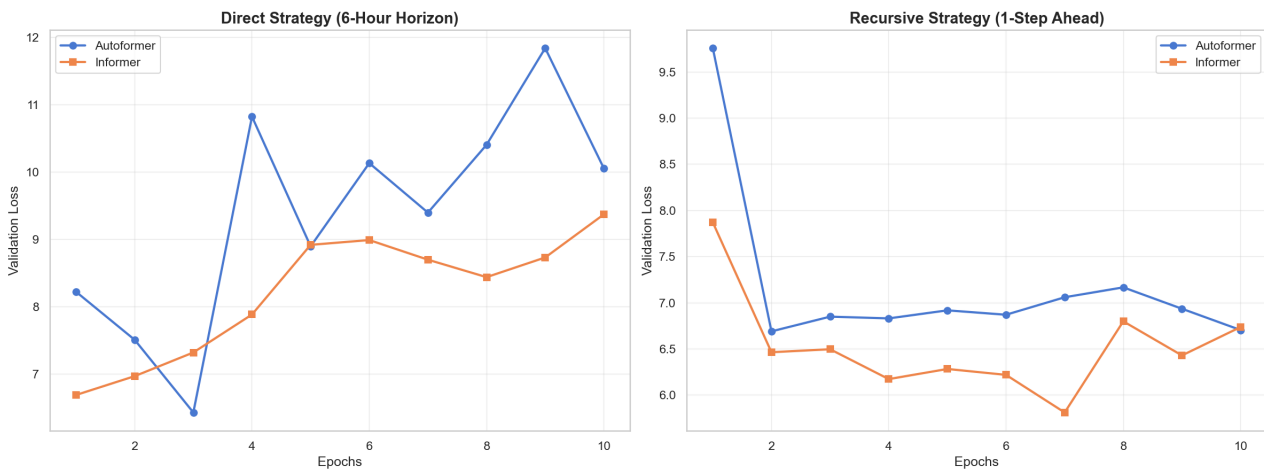
• **sota_robust/best_performance_comparison.png**



• **sota_robust/convergence_analysis.png**

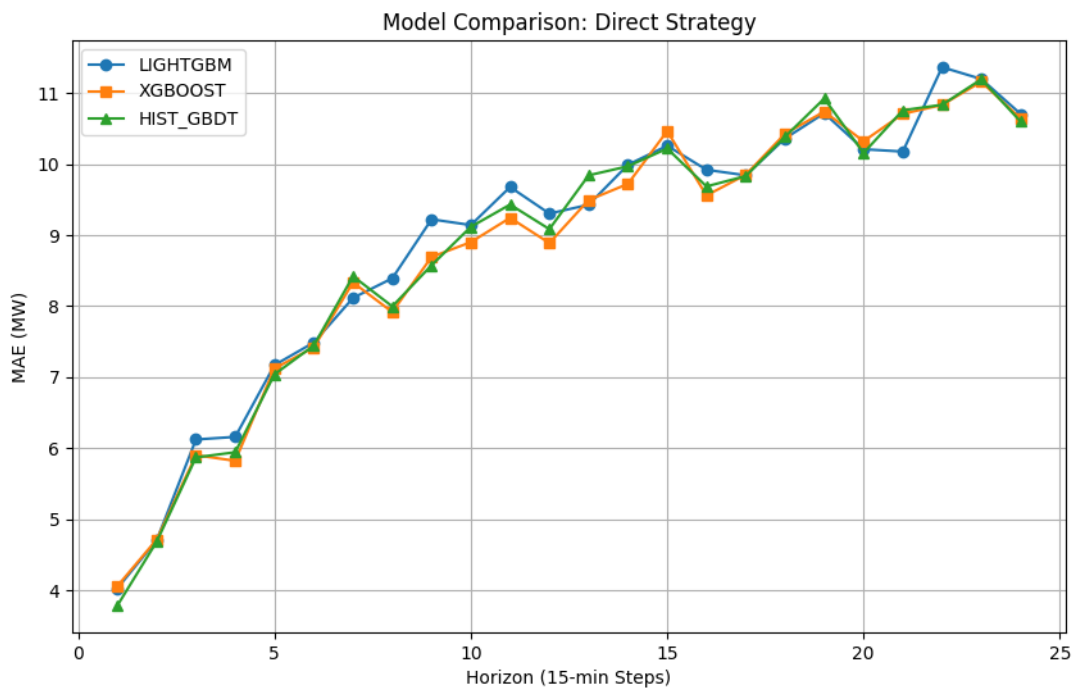


• **sota_robust/strategy_comparison.png**

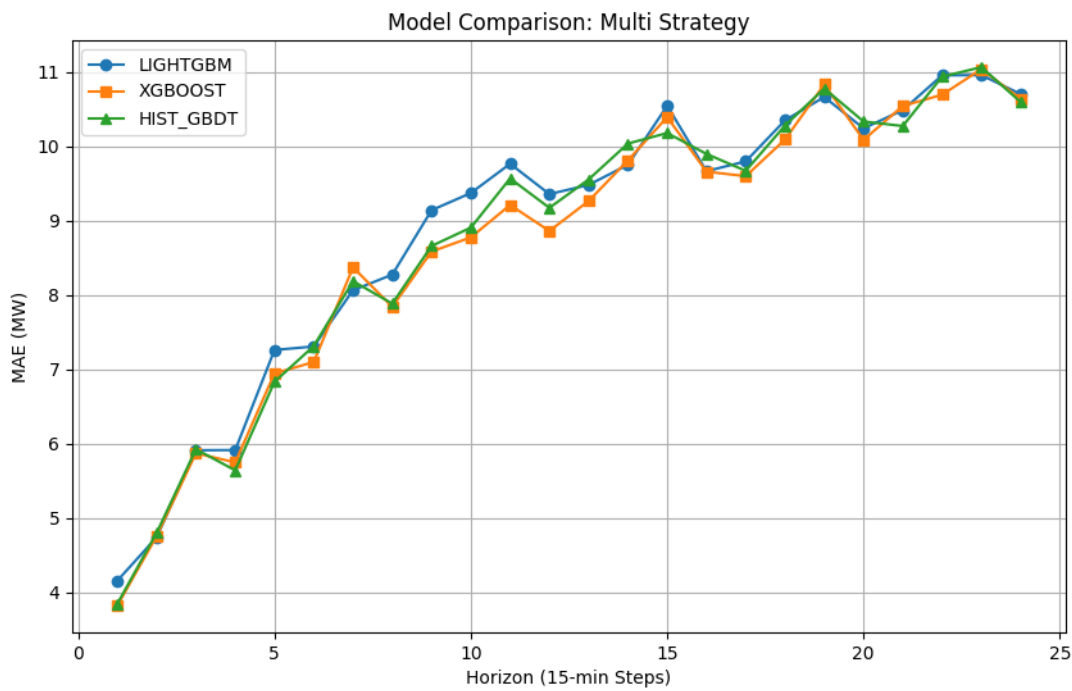


Strategies Comparison

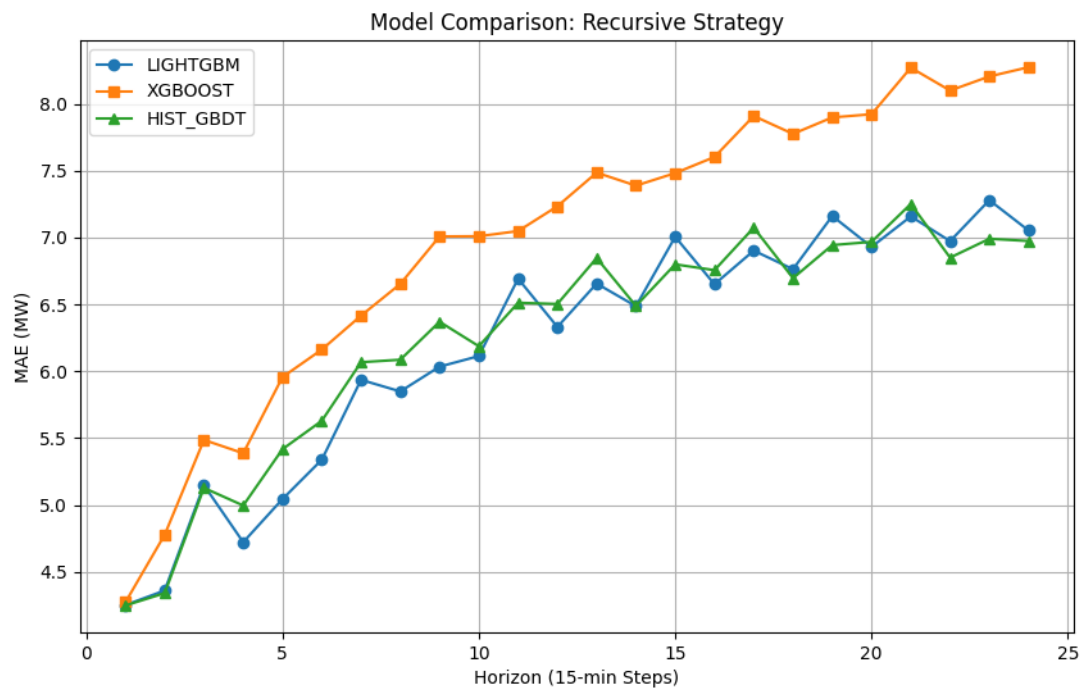
- **strategies/comparison/compare_models_direct.png**



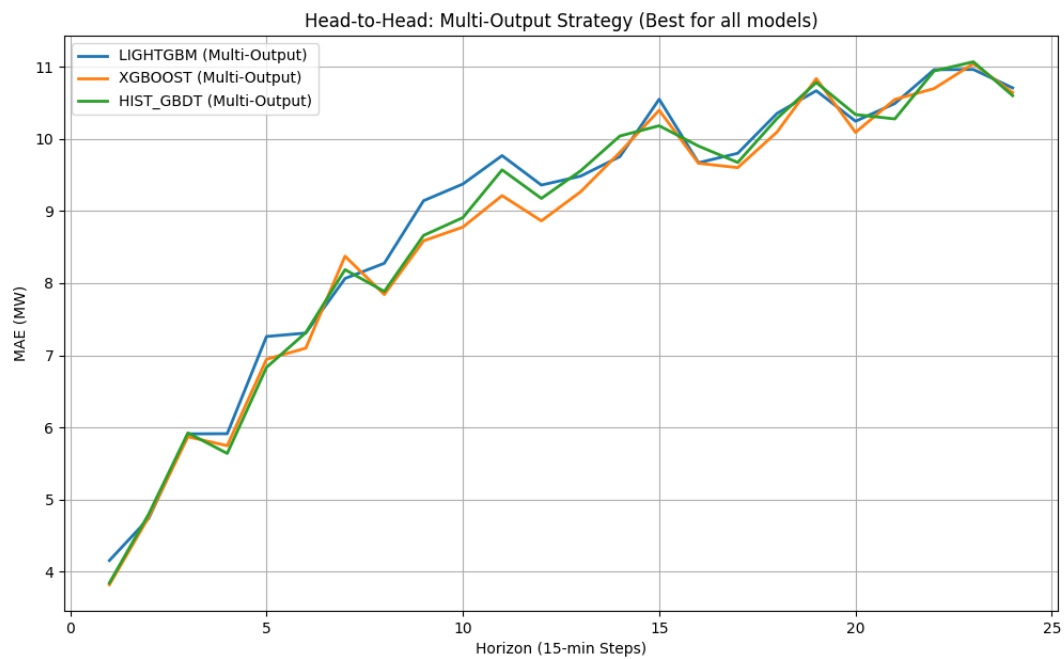
- **strategies/comparison/compare_models_multi.png**



- **strategies/comparison/compare_models_recursive.png**

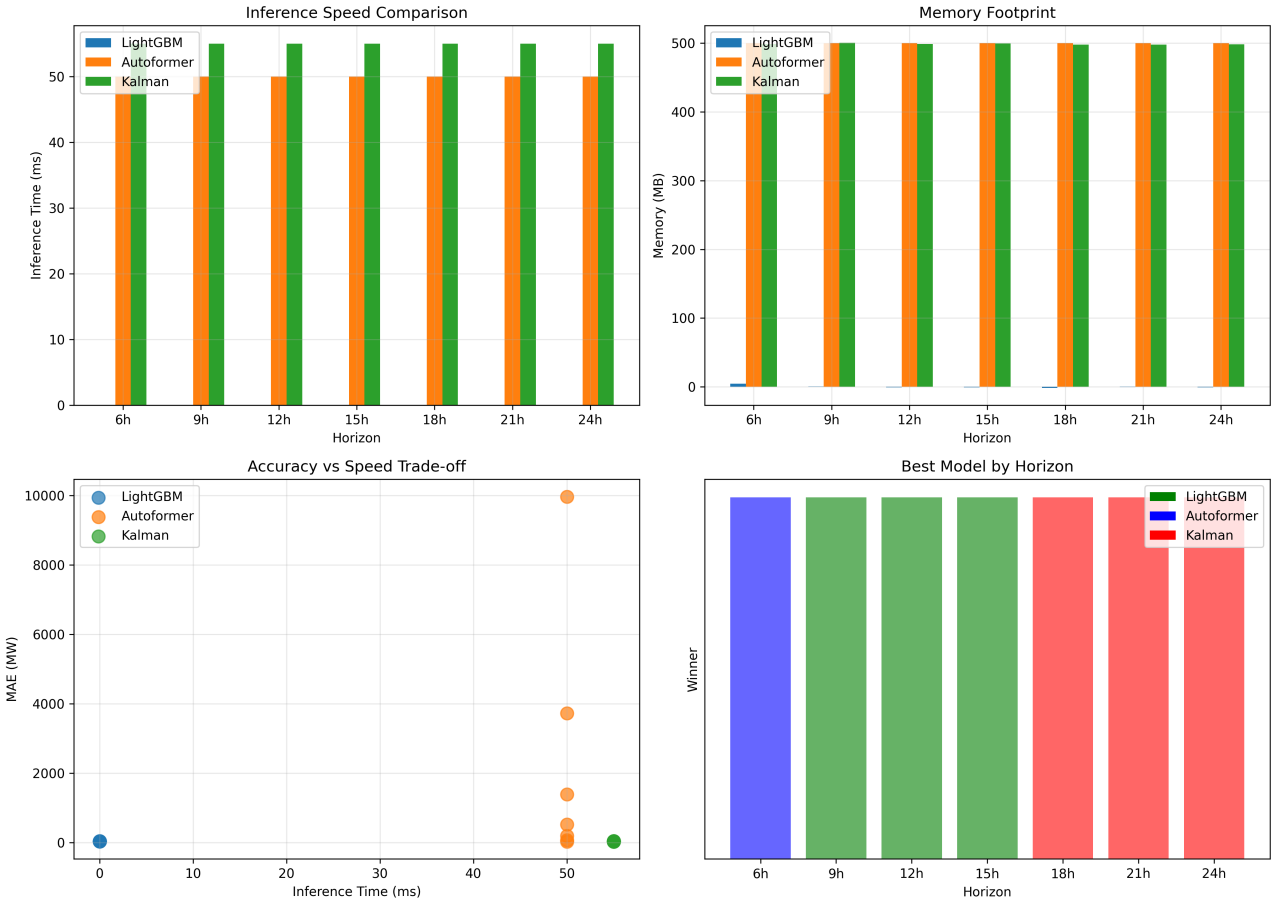


- **strategies/comparison/head_to_head_multi_output.png**



Transition Zone

• transition_zone/cost_analysis.png



• transition_zone/mae_growth_profile.png

